

Metamaterial Absorber for RF Energy Harvesting: A design overview

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What are metamaterials?

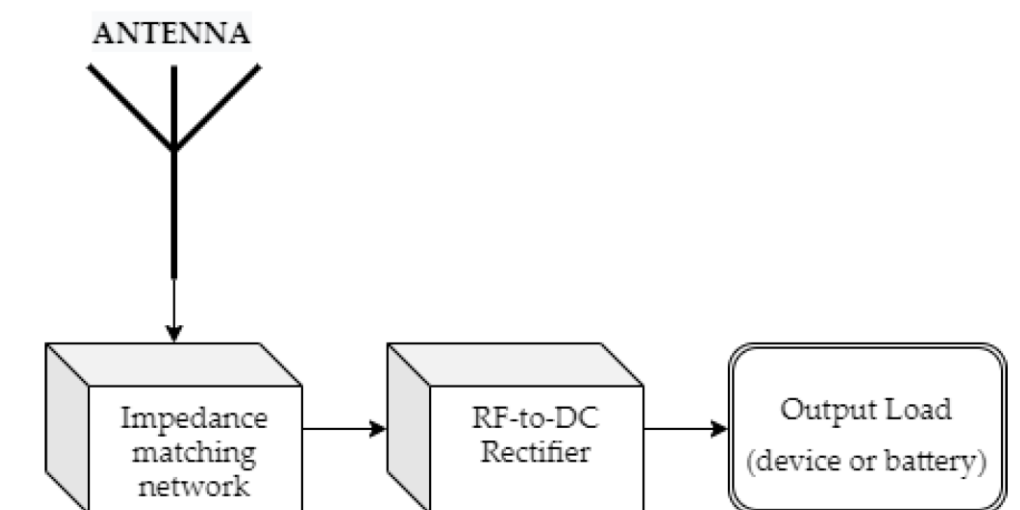
Overview

- Metamaterials are composed of periodic arrays of subwavelength unit cells—structures significantly smaller than the wavelength of the electromagnetic radiation they interact with.
- For RF applications such as EMI shielding and RF energy harvesting, these unit cells are typically constructed from nonmagnetic metals (such as copper) deposited on dielectric substrates like FR-4 or Rogers laminates.
- EM response of these periodic structures differs fundamentally from their constituent materials, giving designers control over material properties
- The material properties are controlled through the unique geometric design of copper that sits on top of the dielectric.
- Thus, they control two parameters: **Permittivity (ϵ)** - the material's response to the electric field component of electromagnetic waves.
Permeability (μ) - the material's response to the magnetic field component of electromagnetic waves
- Response can be couple electrically or magnetically
- Placement in an array promotes mutual coupling, which increases harvest from cascading amplification

Metamaterials continued

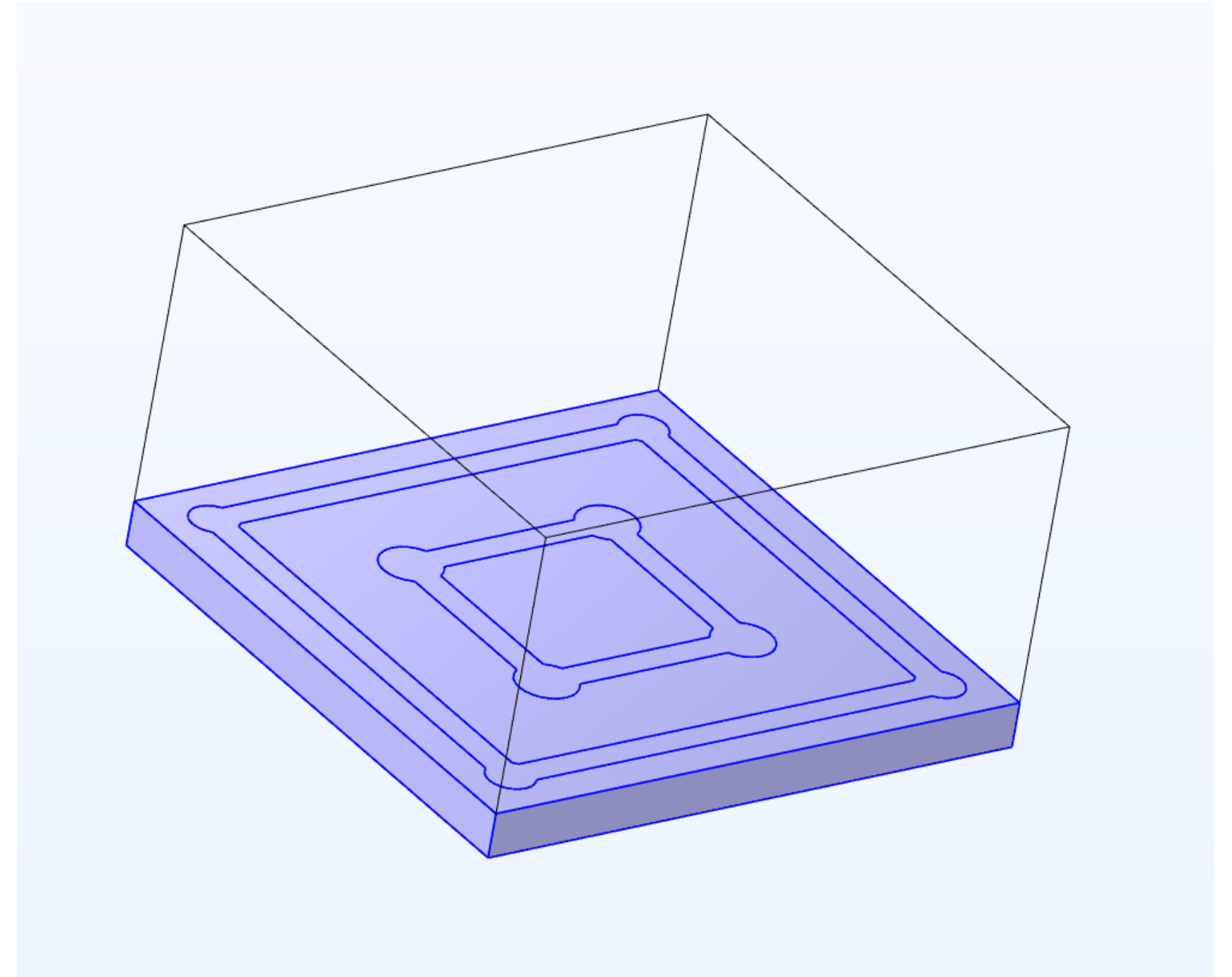
Overview

- Metamaterials function in our context as frequency-selective metamaterial absorbers (MA); structures engineered to absorb electromagnetic energy, not reflect or transmit
- Strategic placement of gaps and closed loops creates both electric resonances (from gaps) and magnetic resonances (from closed loops), which can be tuned to specific frequency bands
- The effective impedance of the metamaterial is matched to free space impedance, minimizing reflections and maximizing energy absorption
- Absorption can be either be ohmic losses in the form of heat, or electric current circulation
- Traditional MMAs are more concerned with frequency-selective total absorption, while energy harvesting devices rely on the electric current generation efficiency from total absorption
- RF electromagnetic waves → AC voltage/current → Diode rectification → DC voltage/current (path for energy harvesting)



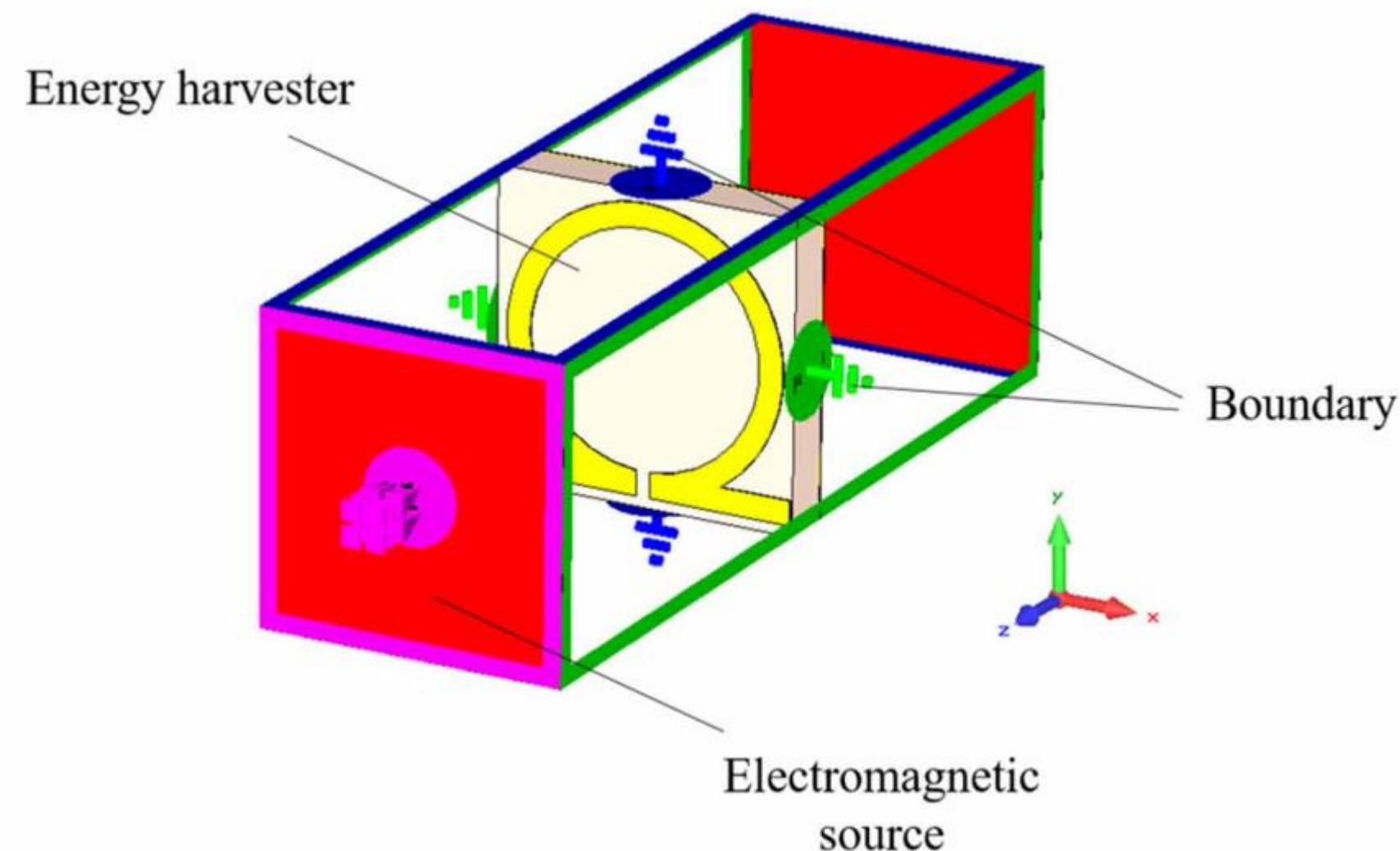
Design

- A low-cost, polarization-independent, wide-angle metamaterial absorber
- 1 oz copper (one resonant structure, one ground plane)
- 1.6 FR4 dielectric (relative permittivity ~ 4.4 , loss tangent ~ 0.020)
- Dual Square Ring Resonator for dual band absorption (2.45 GHz, and 5.7 GHz)
- Simulation techniques inspired by:
<https://www.comsol.com/video/simulating-metamaterials-in-the-terahertz-regime-with-comsol-multiphysics>



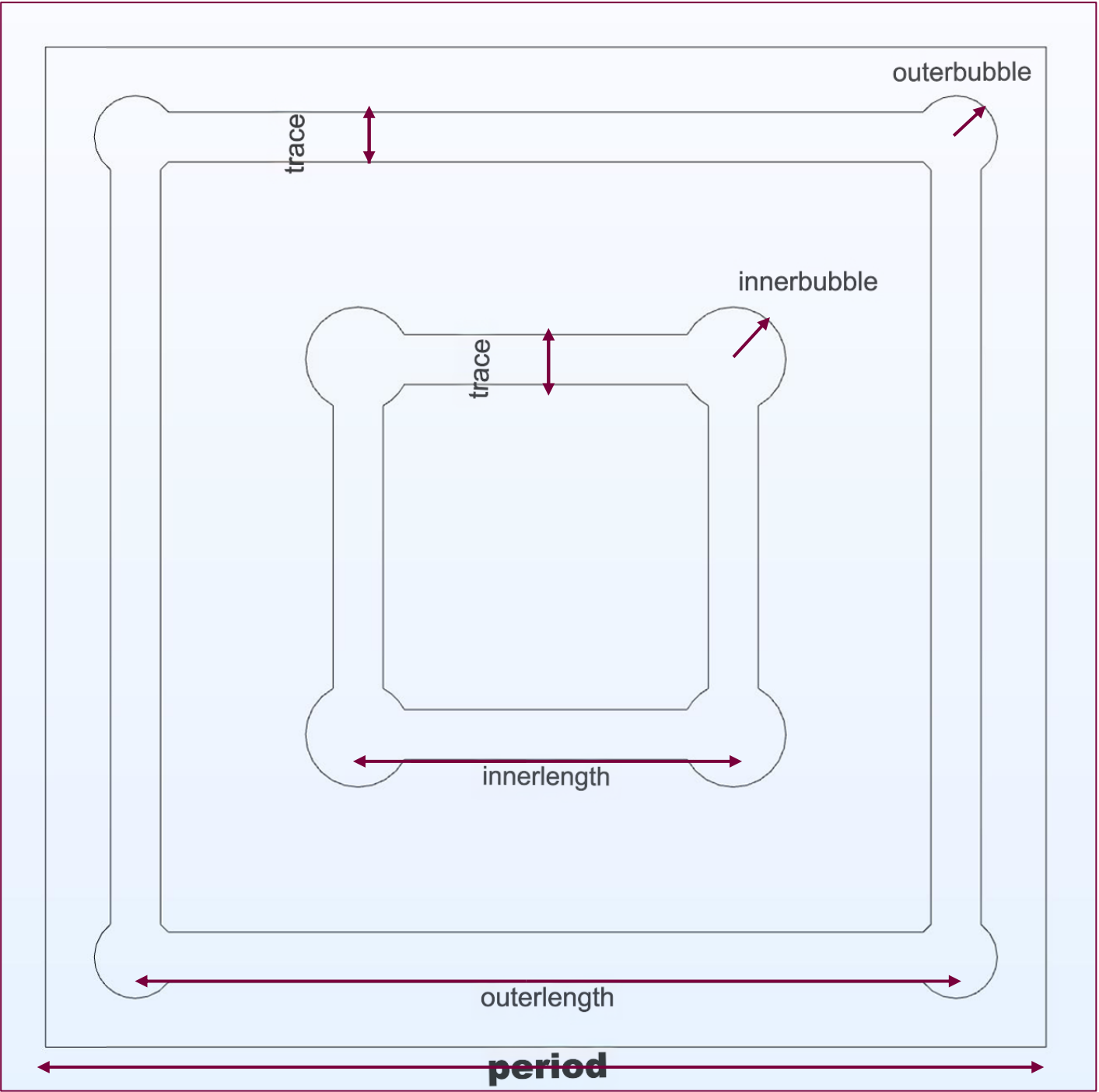
Simulation Inspiration

Design



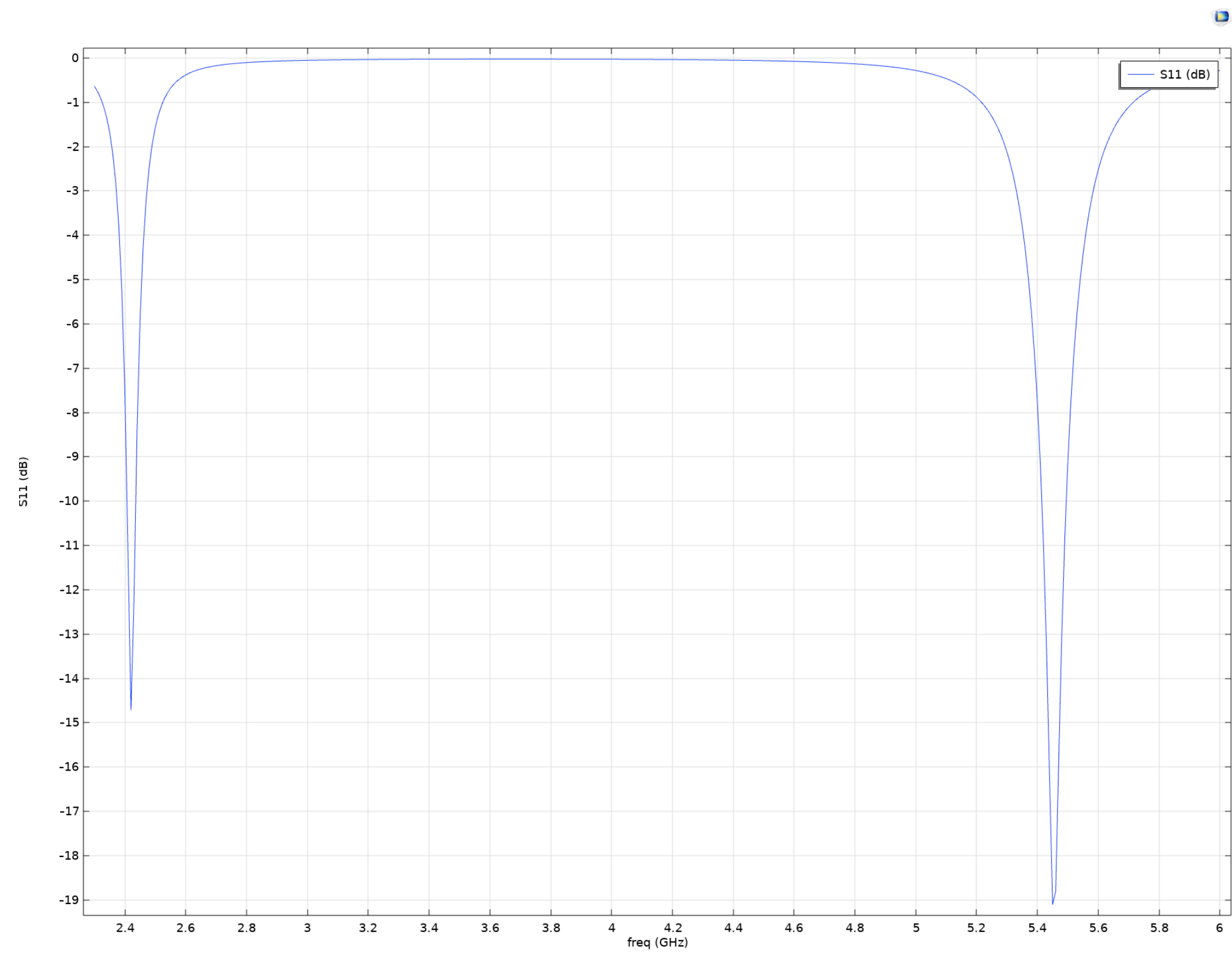
- We set input/output ports (in red) that allow EM energy to enter, which are guided by the alternating PEC and PMC walls shown in blue and green
- To simulate infinite arrays, we set periodic boundary conditions on its walls
- Other valid simulation environments include the use of PML (perfectly matched layer) to simulate an open, free-space boundary and to measure oblique responses. It is however more computationally expensive

Design



Name	Expression	Value	[
period	20[mm]	0.02 m	
dielectric	1.6[mm]	0.0016 m	
trace	1[mm]	0.001 m	
outerlength	17.4[mm]	0.0174 m	
innerlength	8.5[mm]	0.0085 m	
outerbub...	0.825[mm]	8.25E-4 m	
innerbub...	1.05[mm]	0.00105 m	

Design

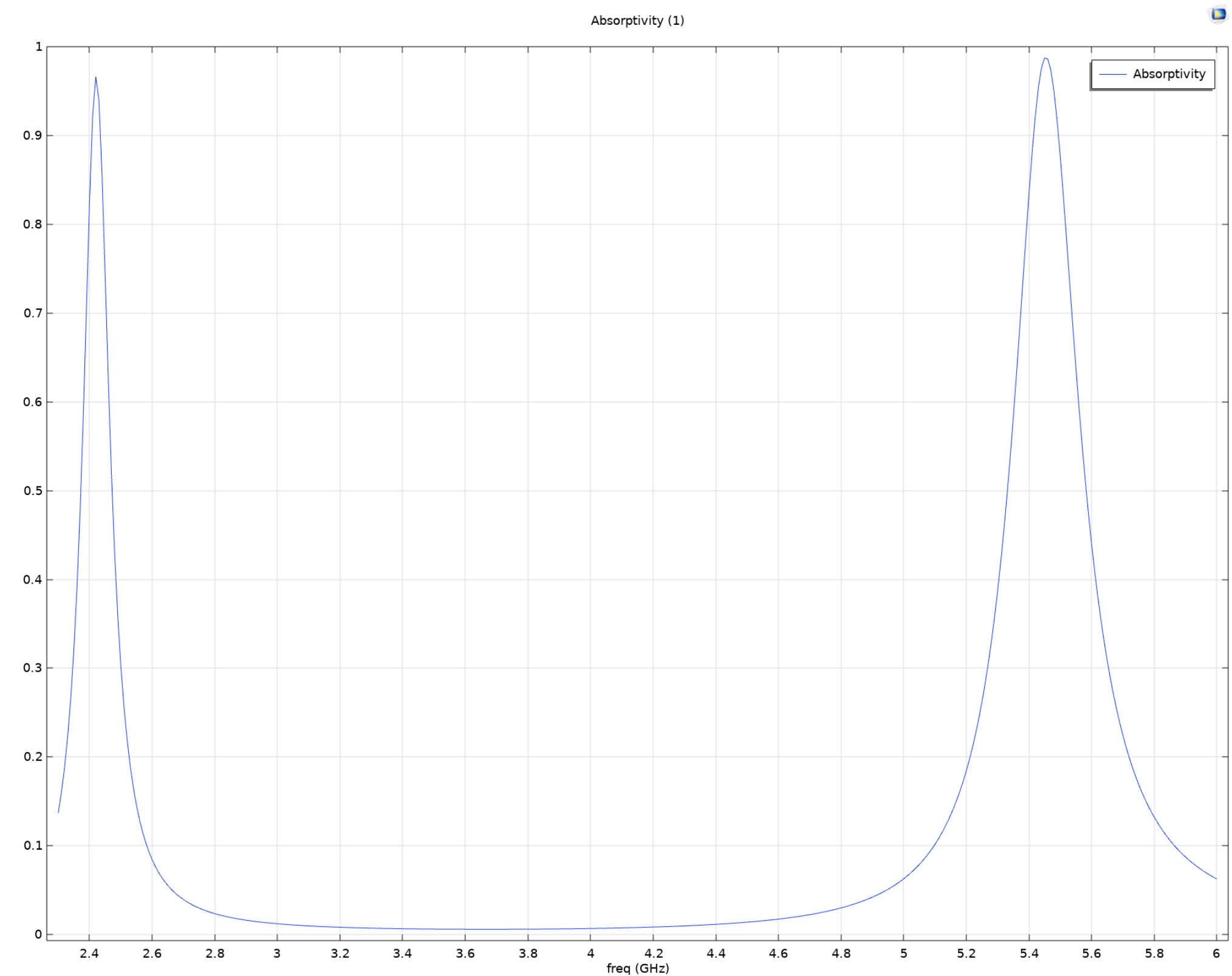


Performance

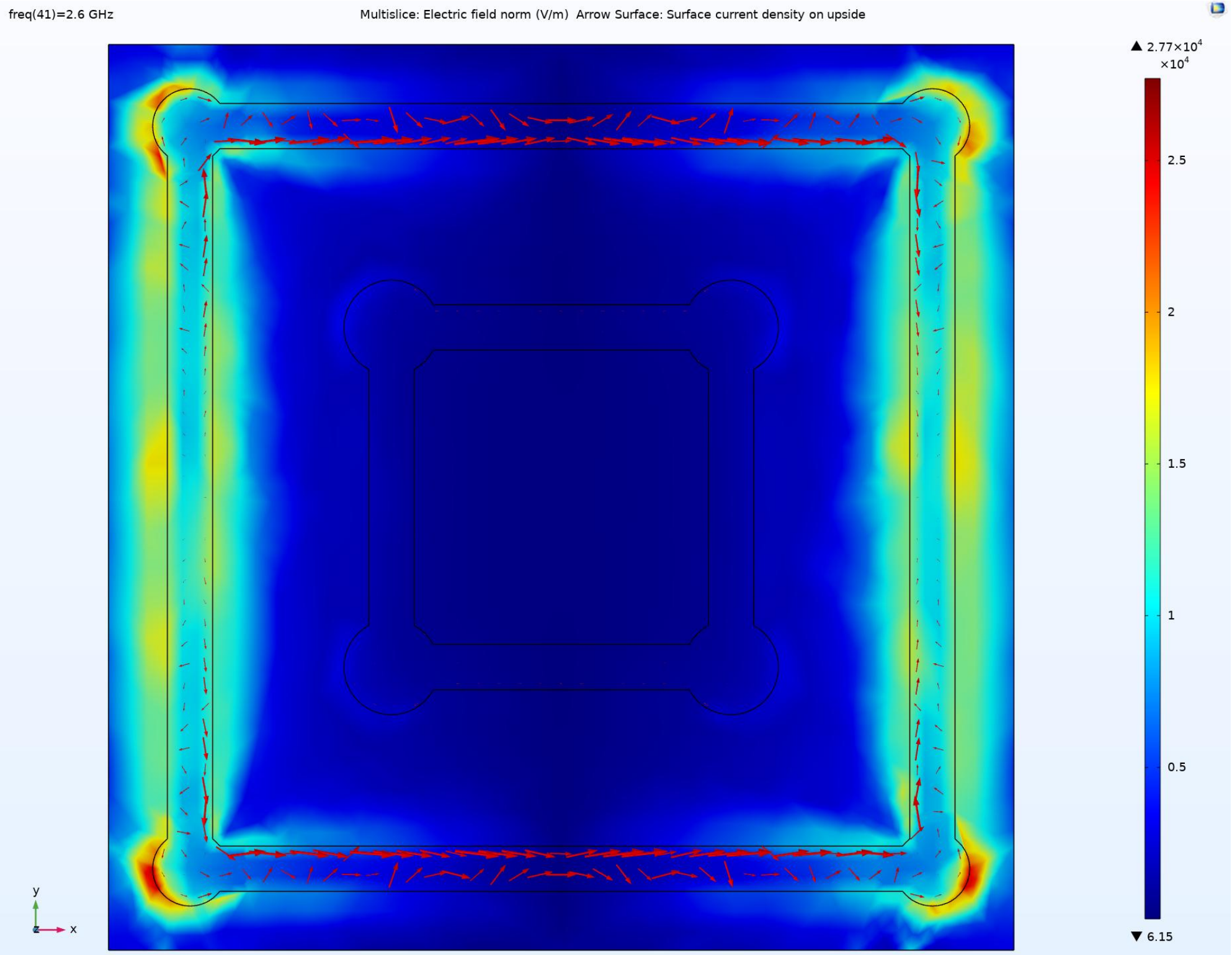
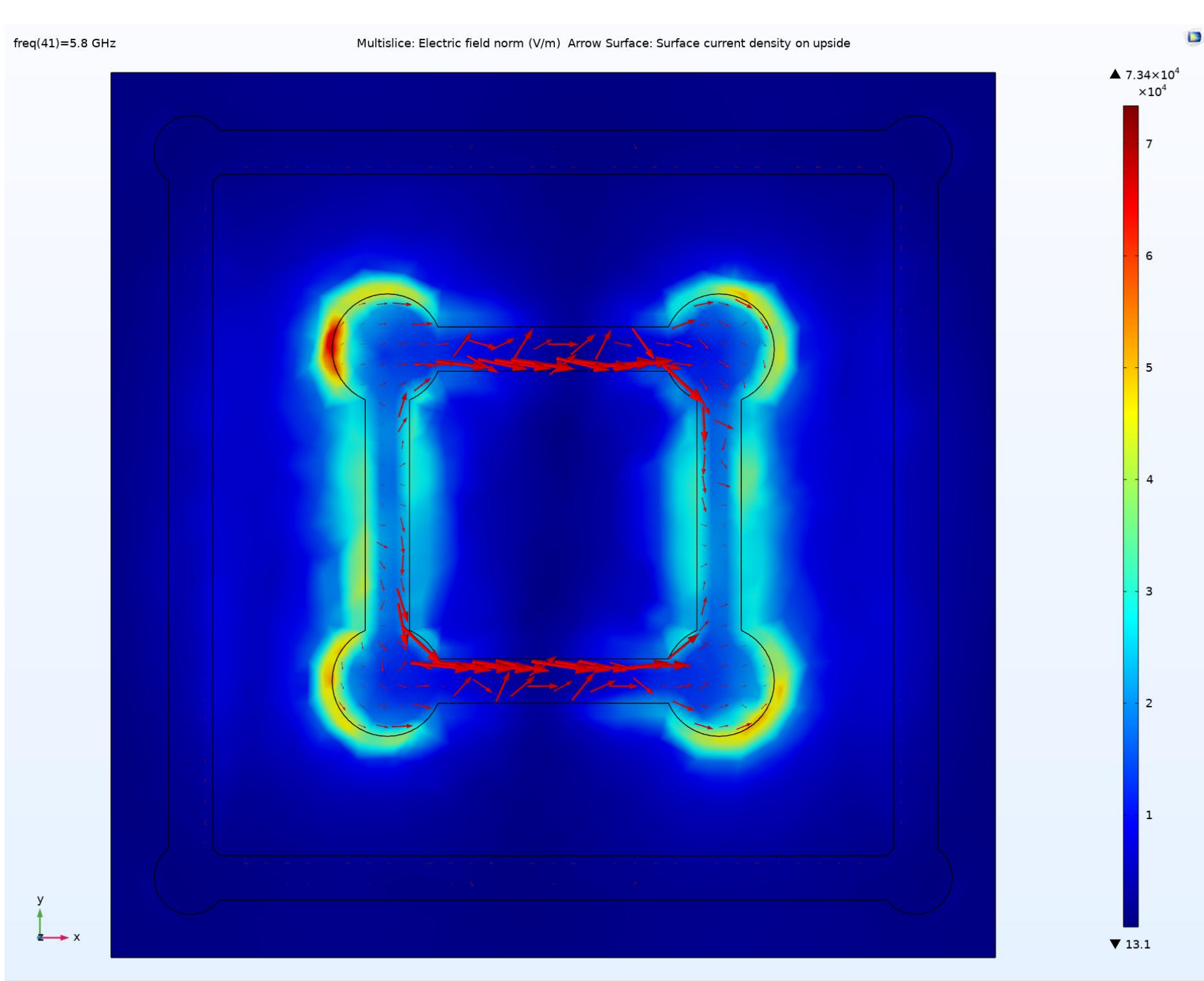
Design

- Outer ring governs 2.4 GHz performance
- Inner ring governs 5.7 GHz performance
- Bubbles on the corners tune the local performance
- Given the copper bottom plane, transmission through the plane is attenuated ($S_{21} = 0$). Reflectivity and Absorptivity are the remaining key parameters
- Input power uses 1 mW

» Expression	Unit	Description
$1 - \text{abs}(\text{emw.S11})^2 - \text{abs}(\text{emw.S21})^2$	1	Absorptivity

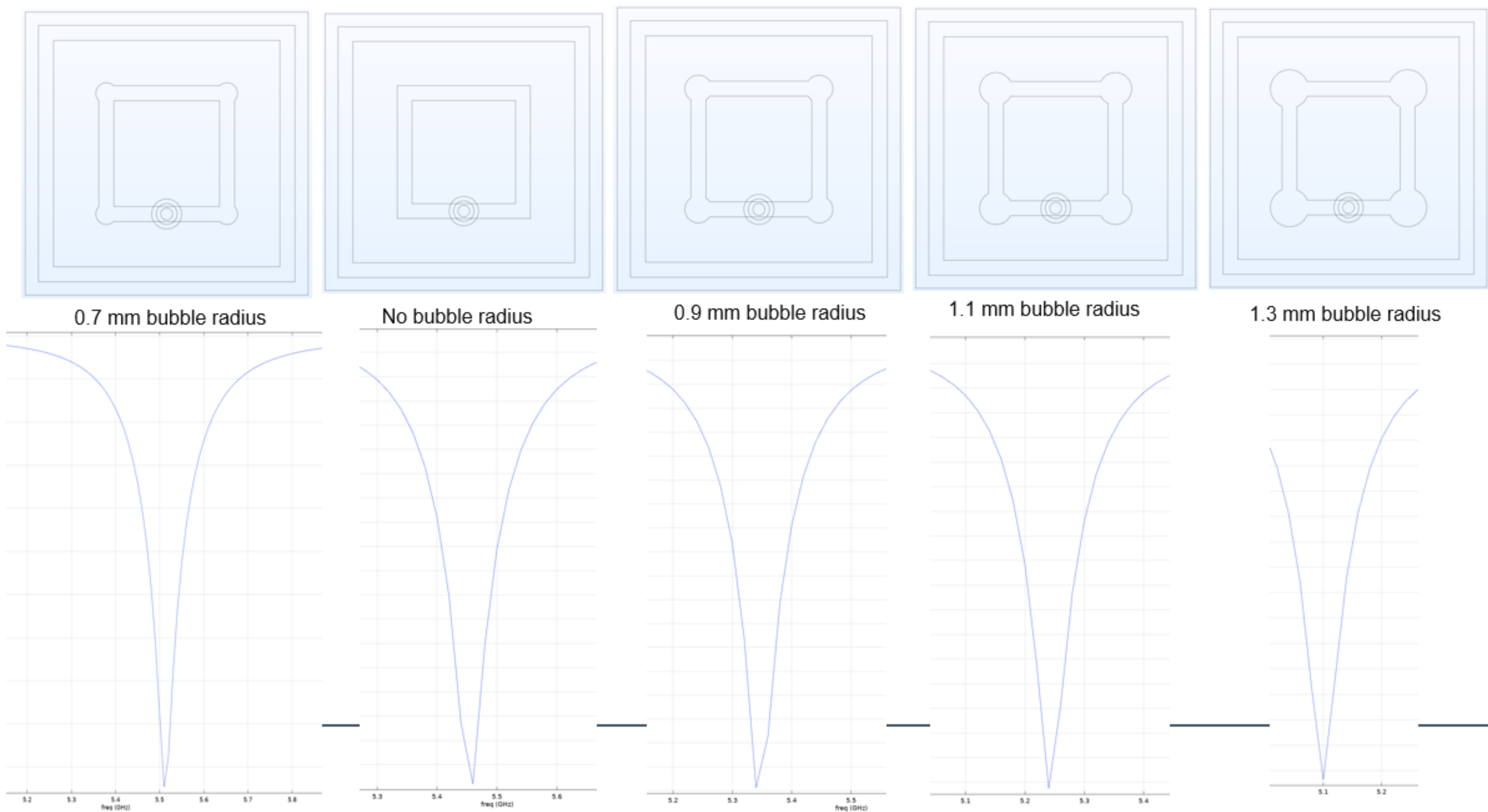


Surface currents



Changing ring sizes (example)

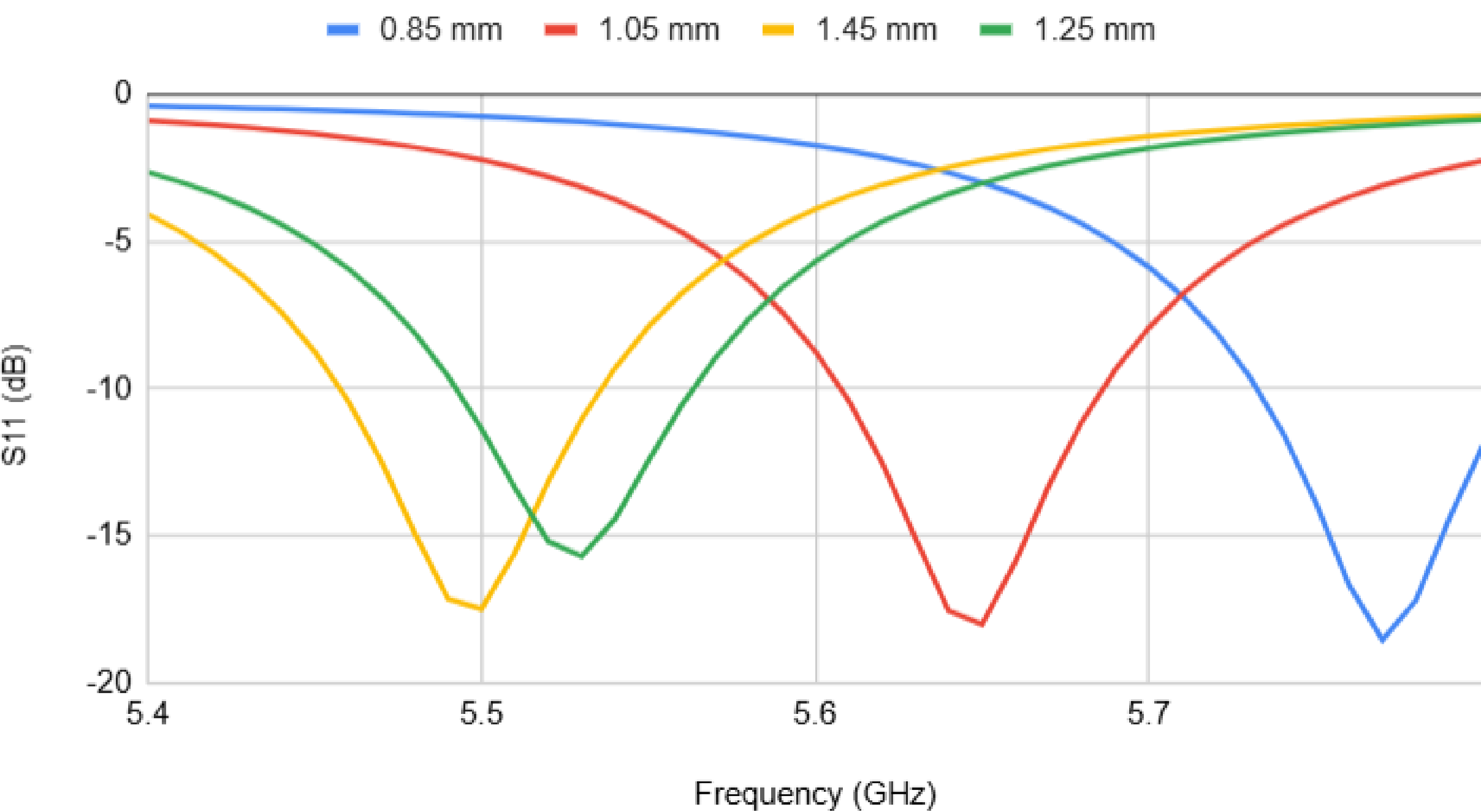
Design



Changing ring sizes (detailed)

Design

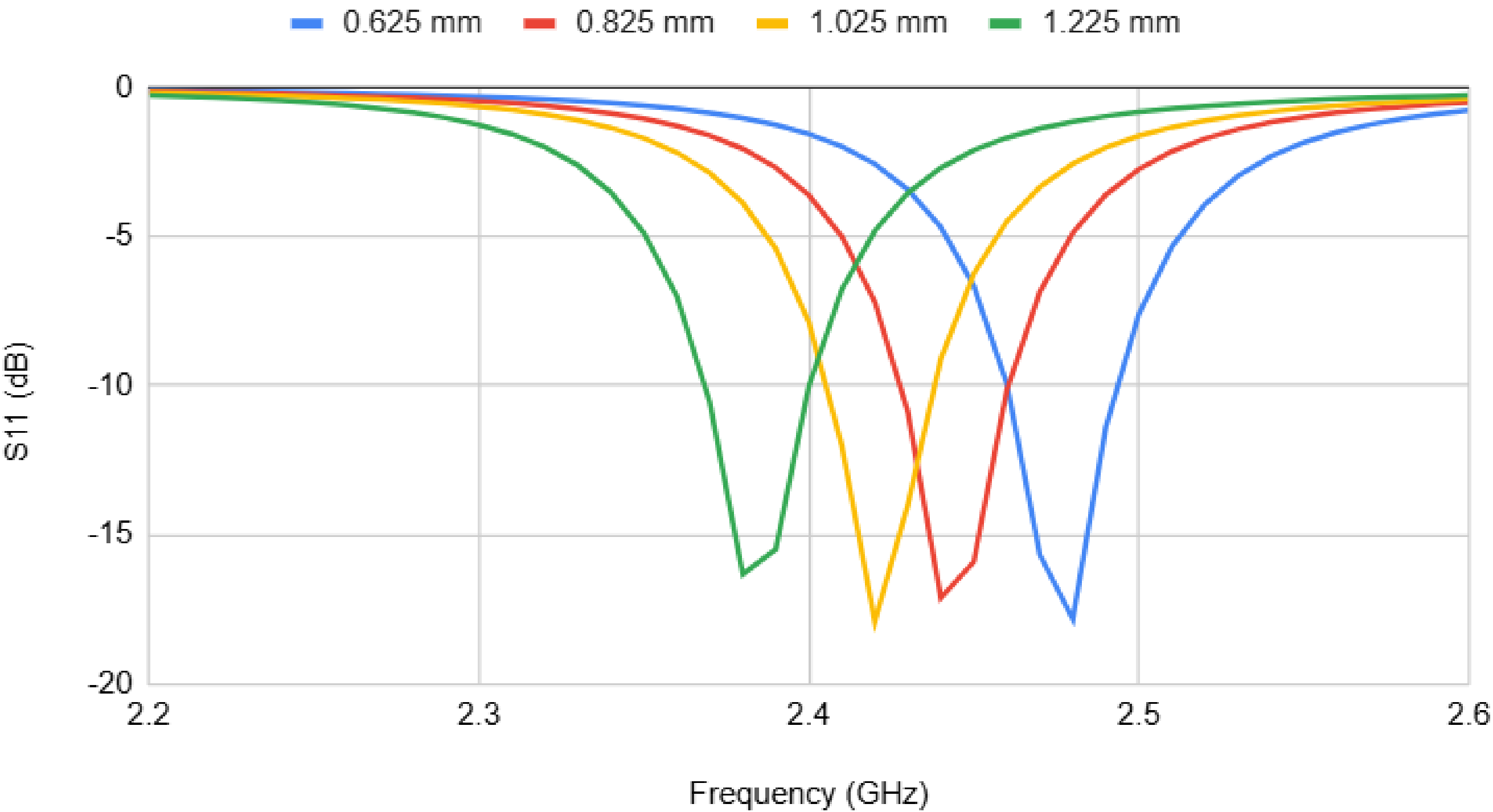
Inner bubble tuning



Changing ring sizes (detailed)

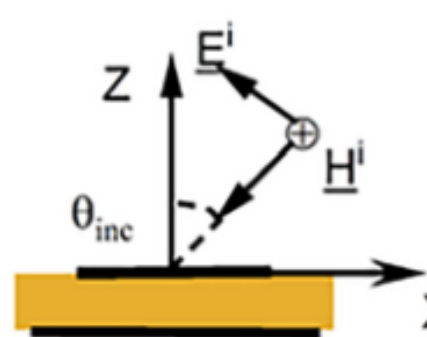
Design

Outer bubble tuning



Changing angle measurements @ 2.4 GHz (from orthogonal to grazing)

Design



TM - $\phi_{inc} = 0^\circ$

Electric mode field:

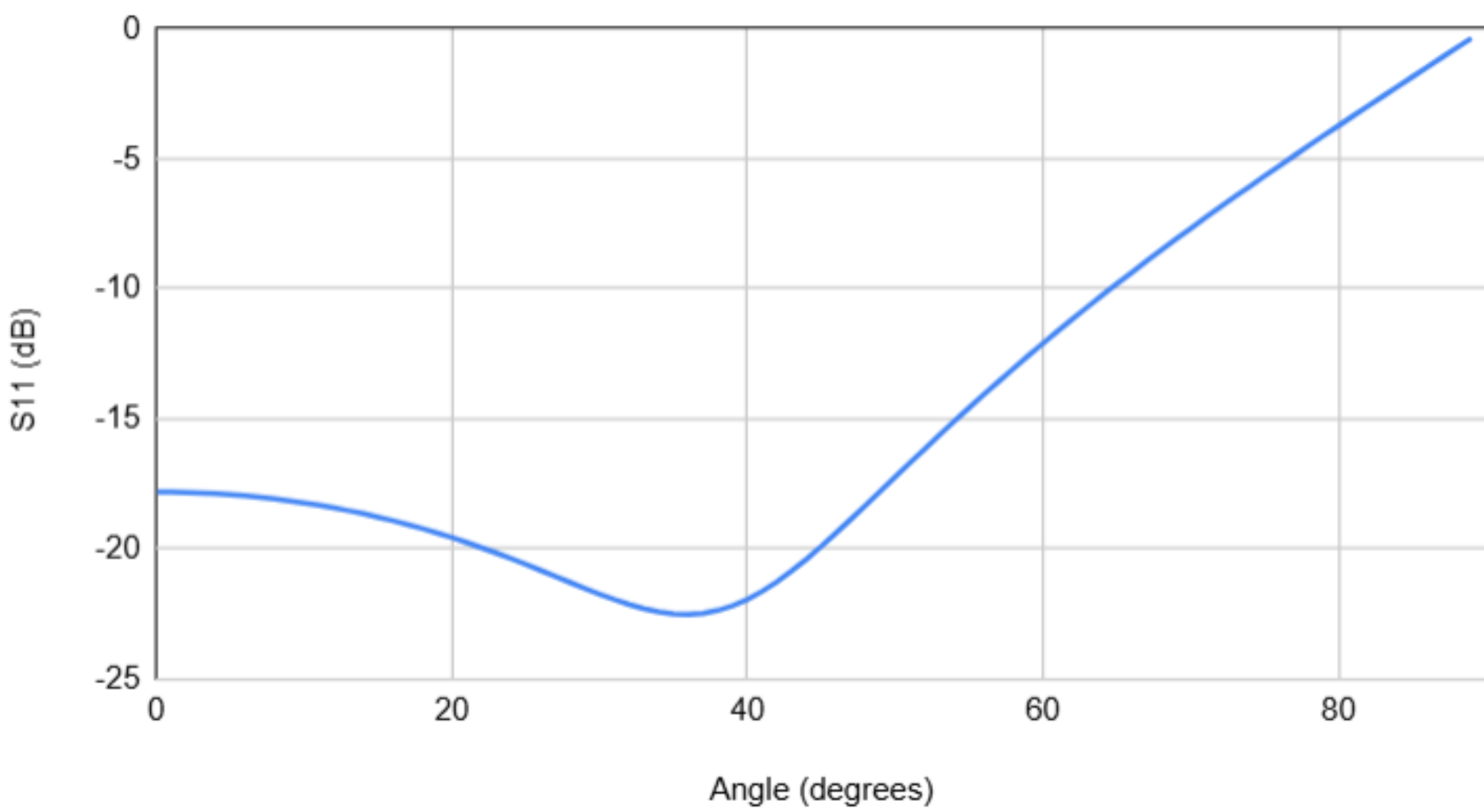
$\cos(\theta) \exp(-j \cdot \text{emw.k} \cdot \sin(\theta) \cdot x)$	x
0	y
$-\sin(\theta) \exp(-j \cdot \text{emw.k} \cdot \sin(\theta) \cdot x)$	z

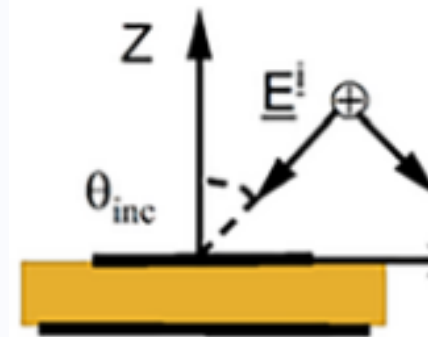
E_0 V/m

Propagation constant:

β	$\text{emw.k} \cdot \cos(\theta)$	rad/m
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TM mode





TE - $\phi_{inc} = 0^\circ$

Magnetic mode field:

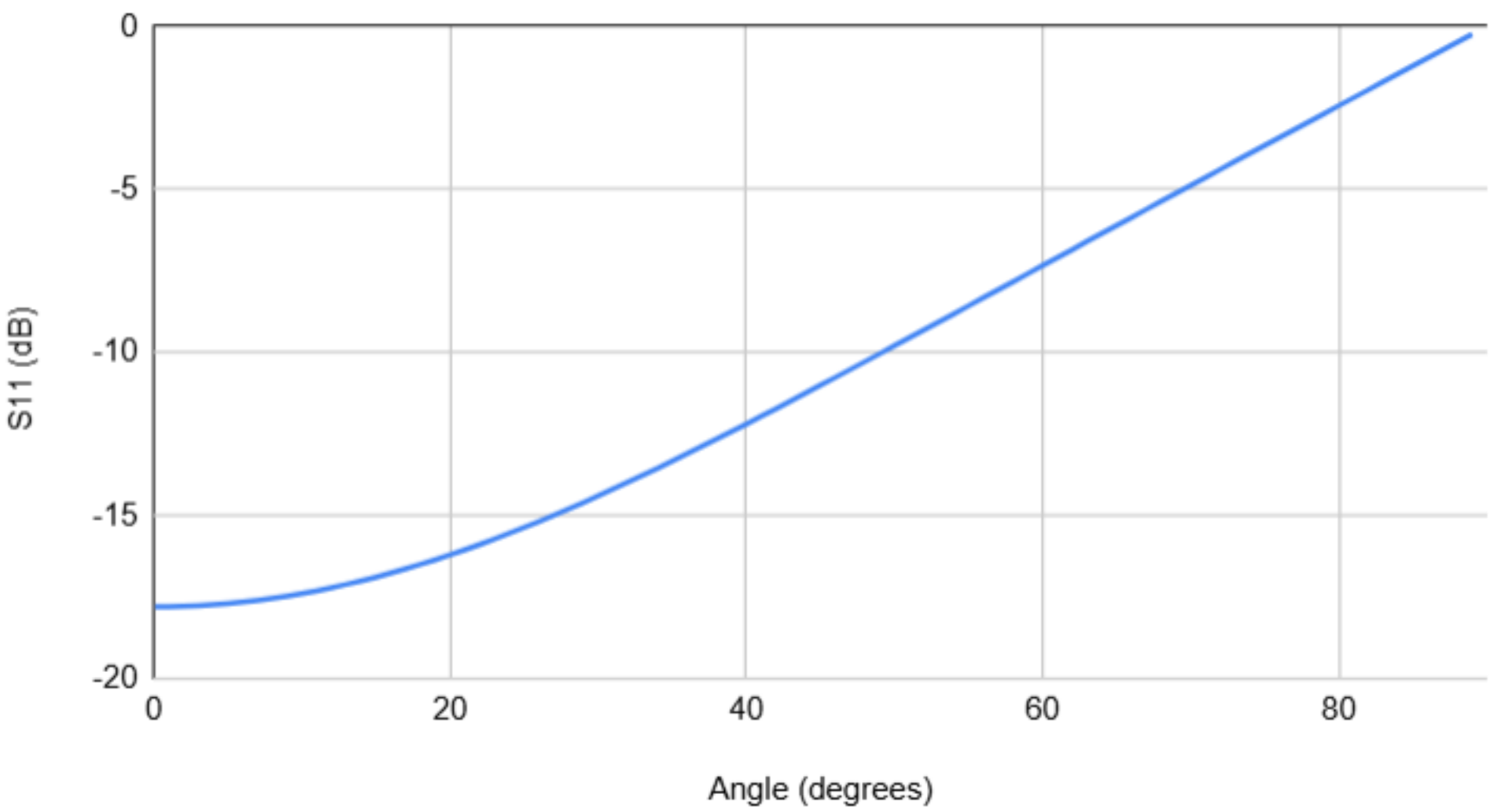
0	x
$\cos(\theta) \exp(-j \cdot \text{emw.k} \cdot \sin(\theta) \cdot x)$	y
$-\sin(\theta) \exp(-j \cdot \text{emw.k} \cdot \sin(\theta) \cdot x)$	z

H_0 A/m

Propagation constant:

β	$\text{emw.k} \cdot \cos(\theta)$	rad/m
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TE mode

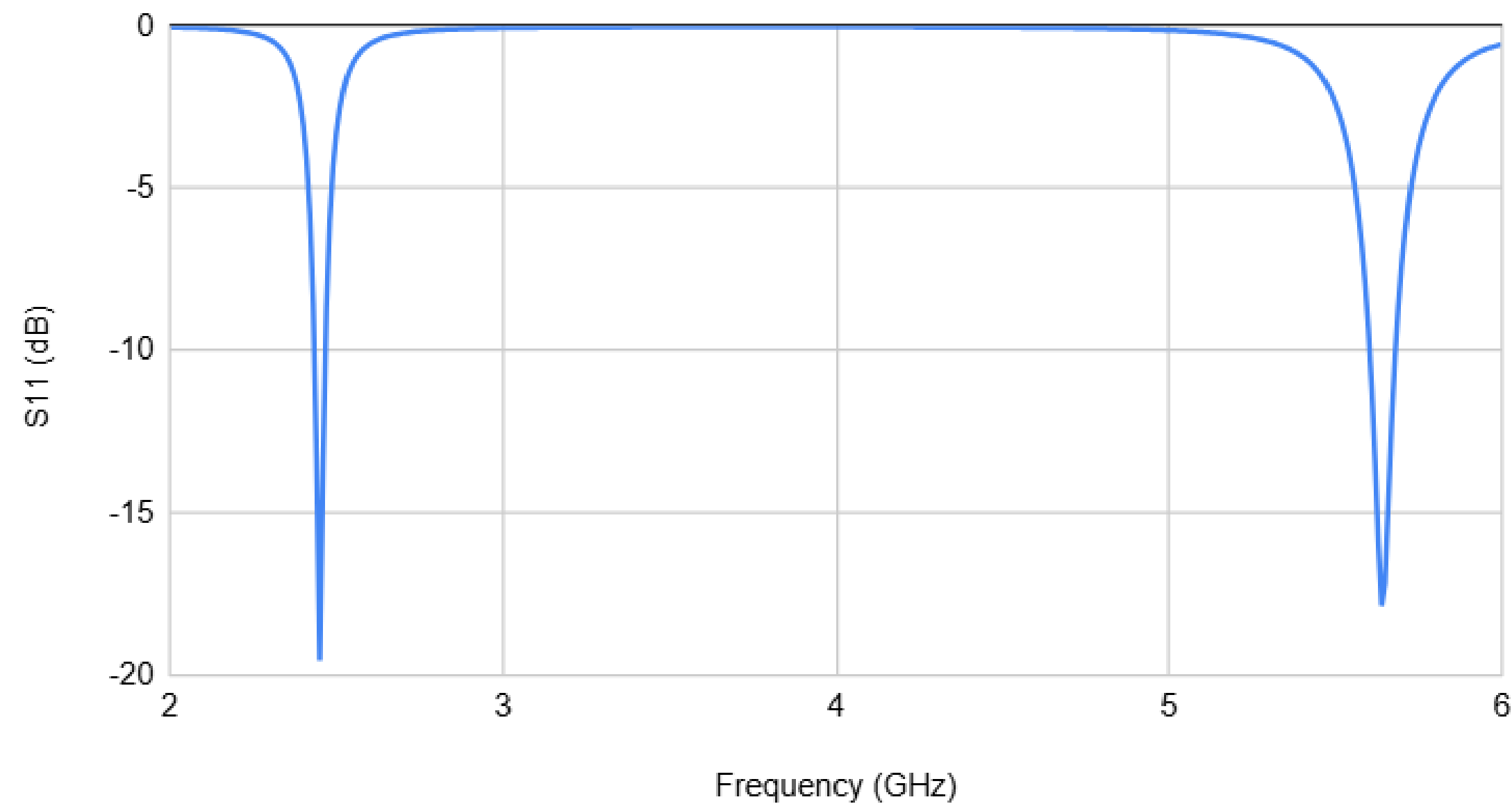


From unit cell to 8x8

Design

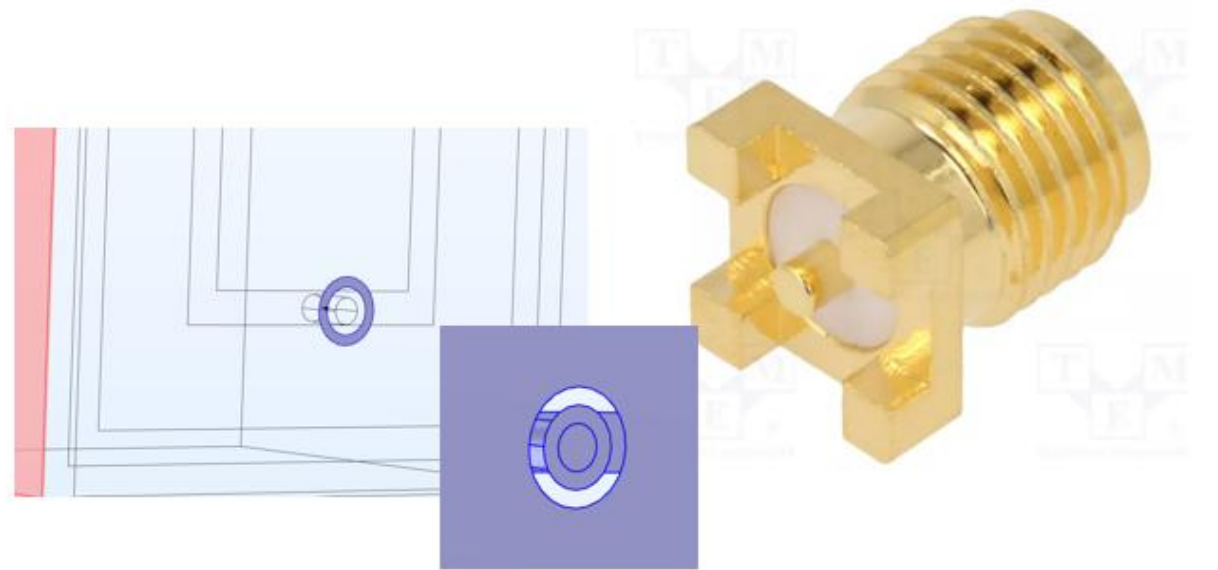
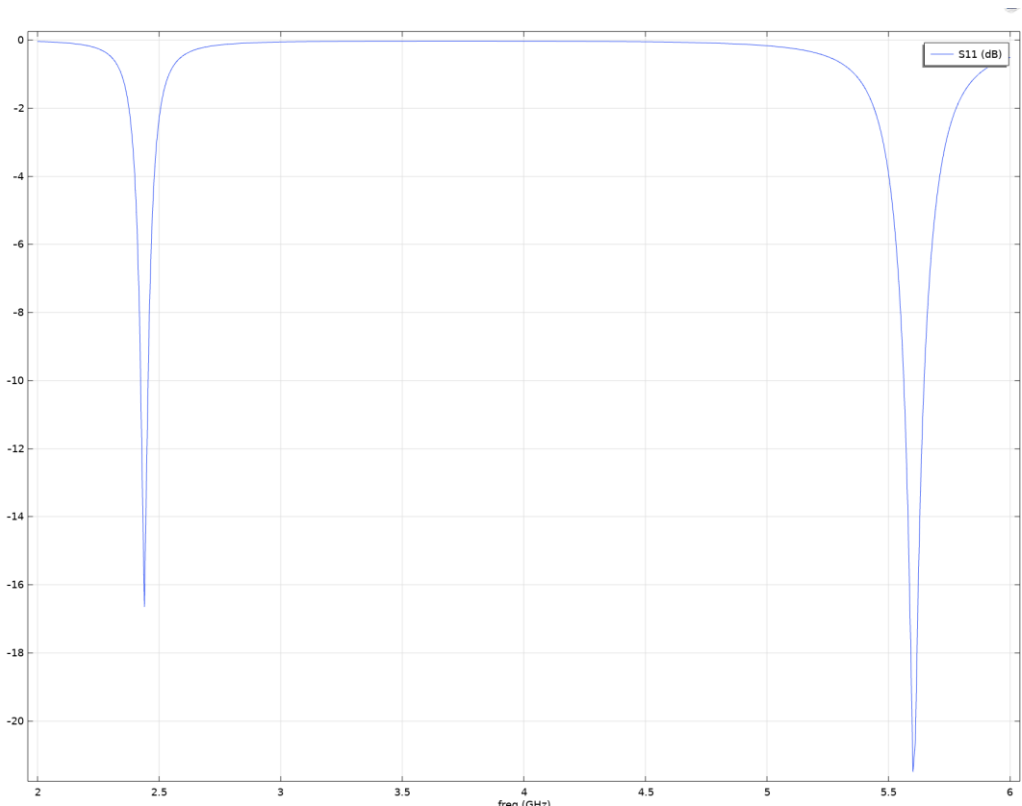
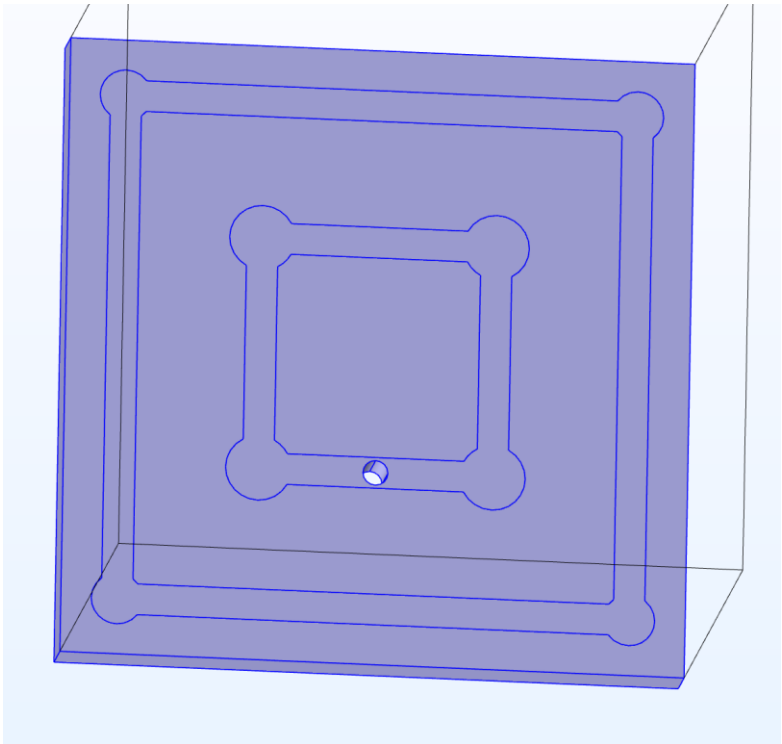
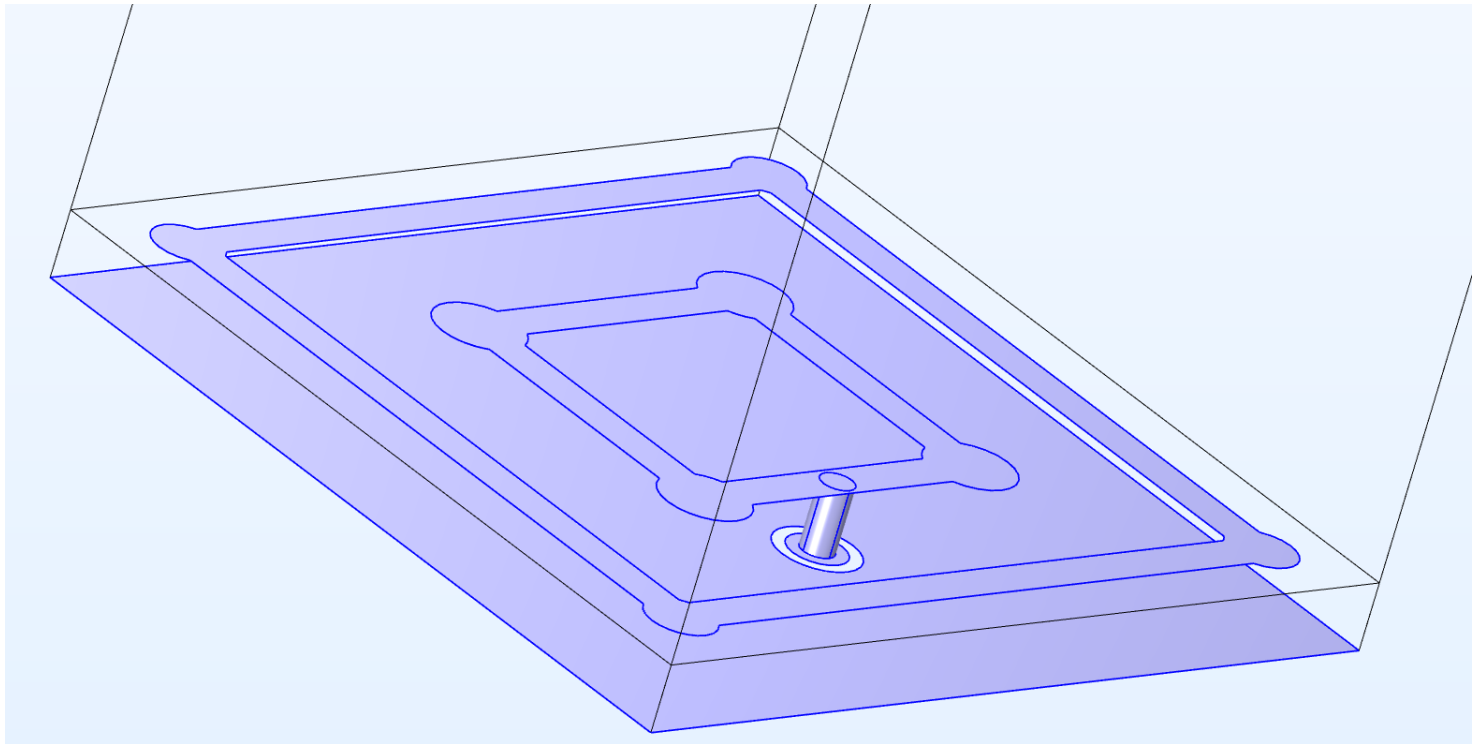
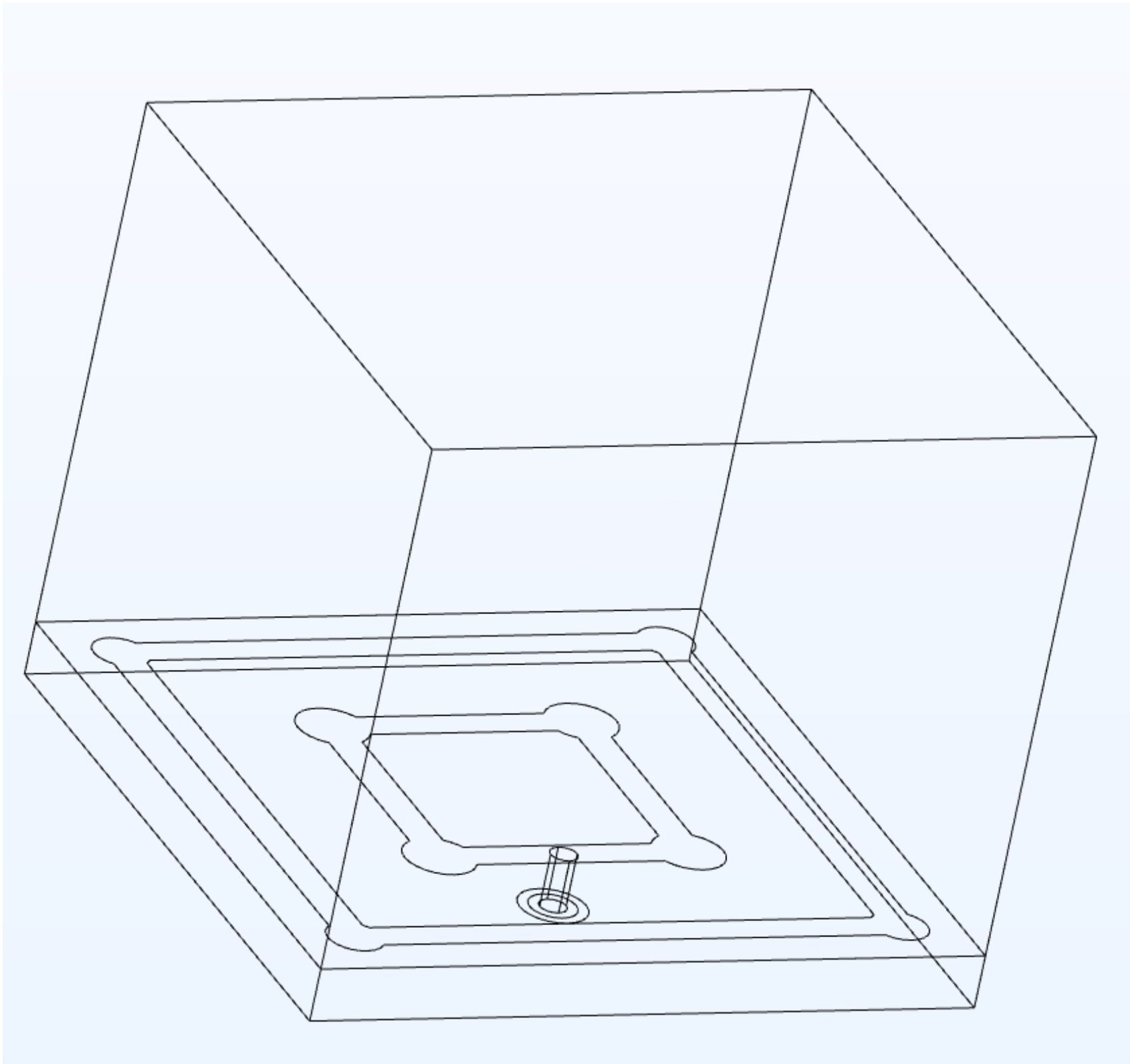
- Turning unit cell geometry into an 8x8 model

8x8 metamaterial



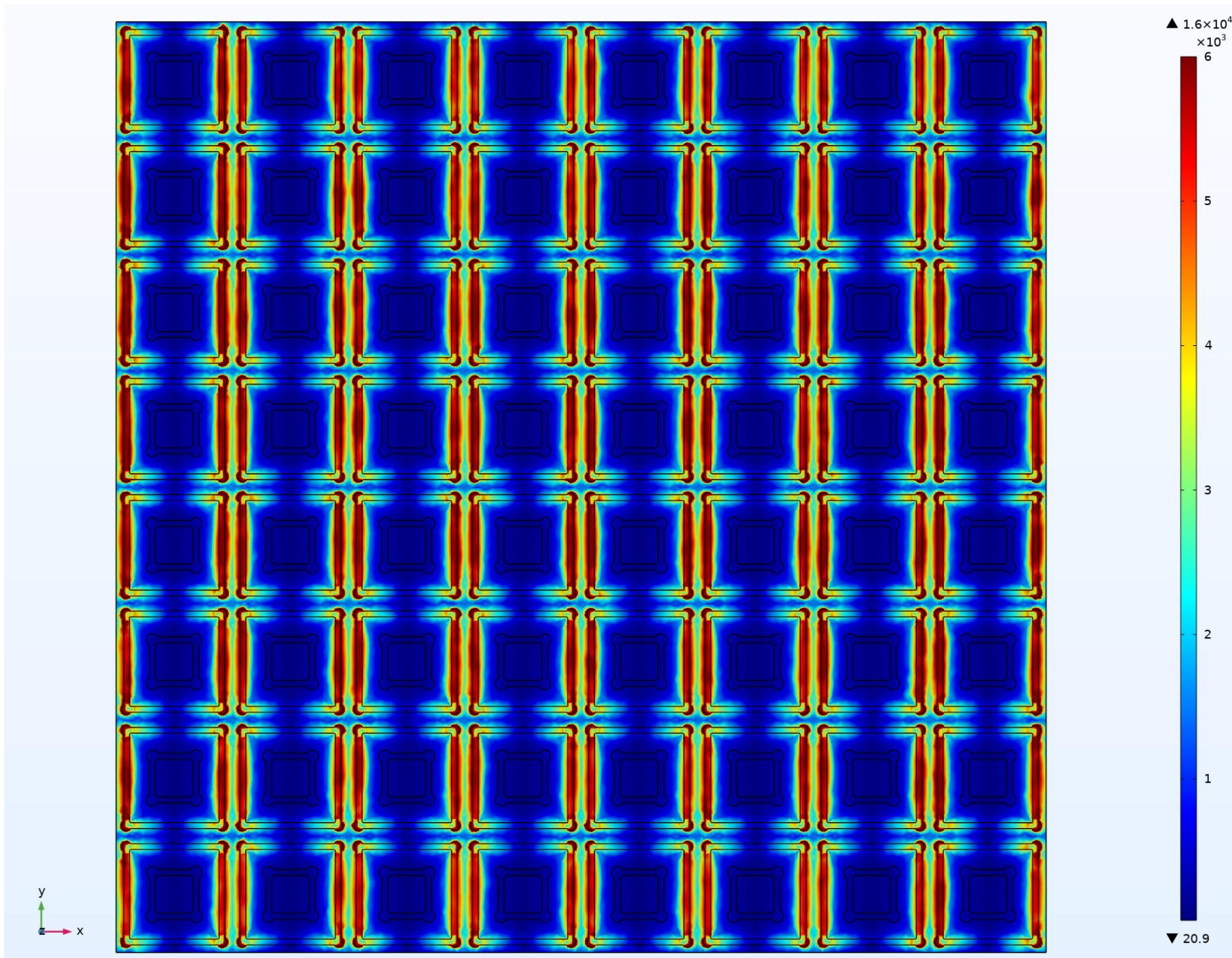
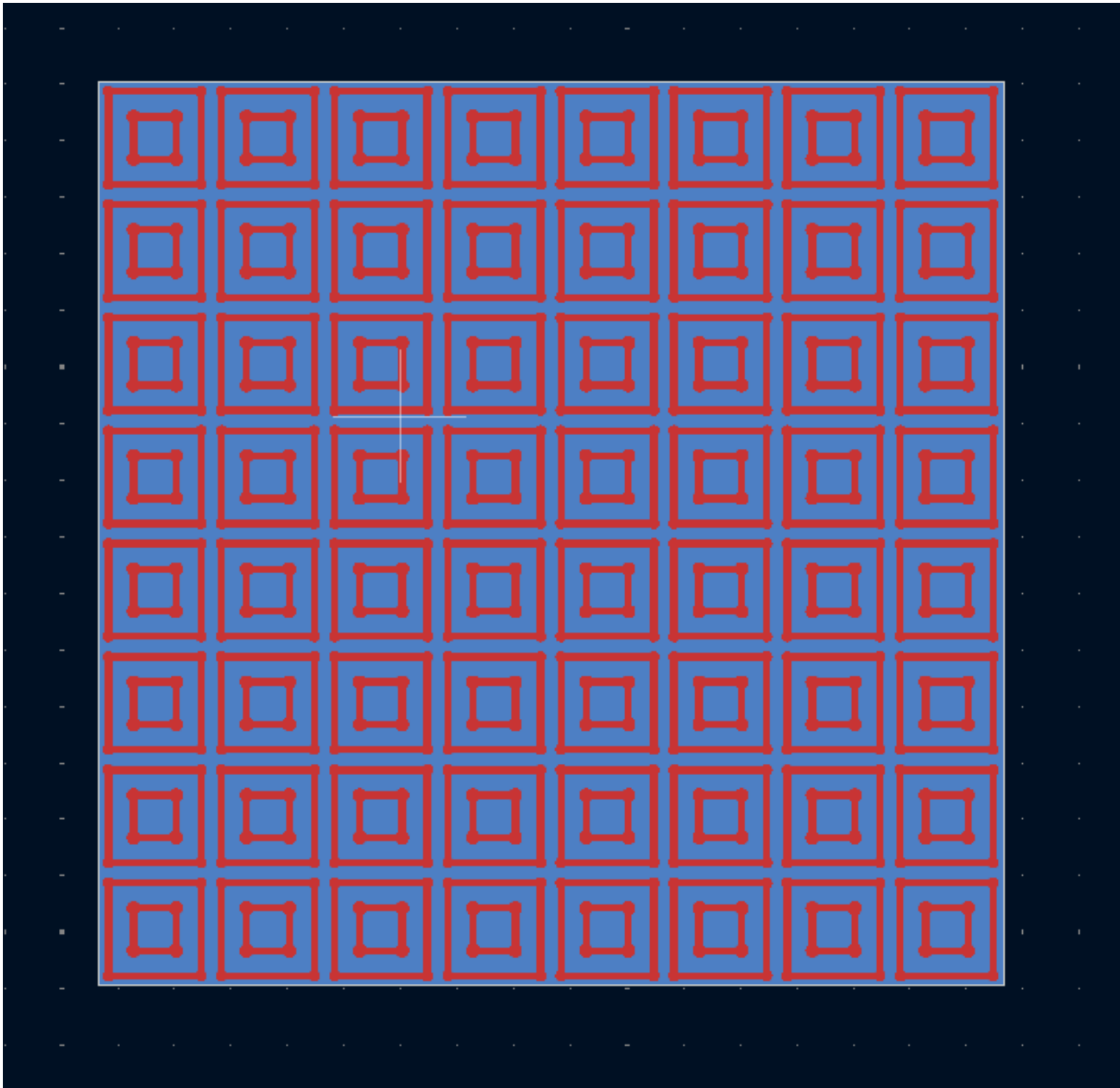
Inclusion of via for harvesting (Simulation only)

Design



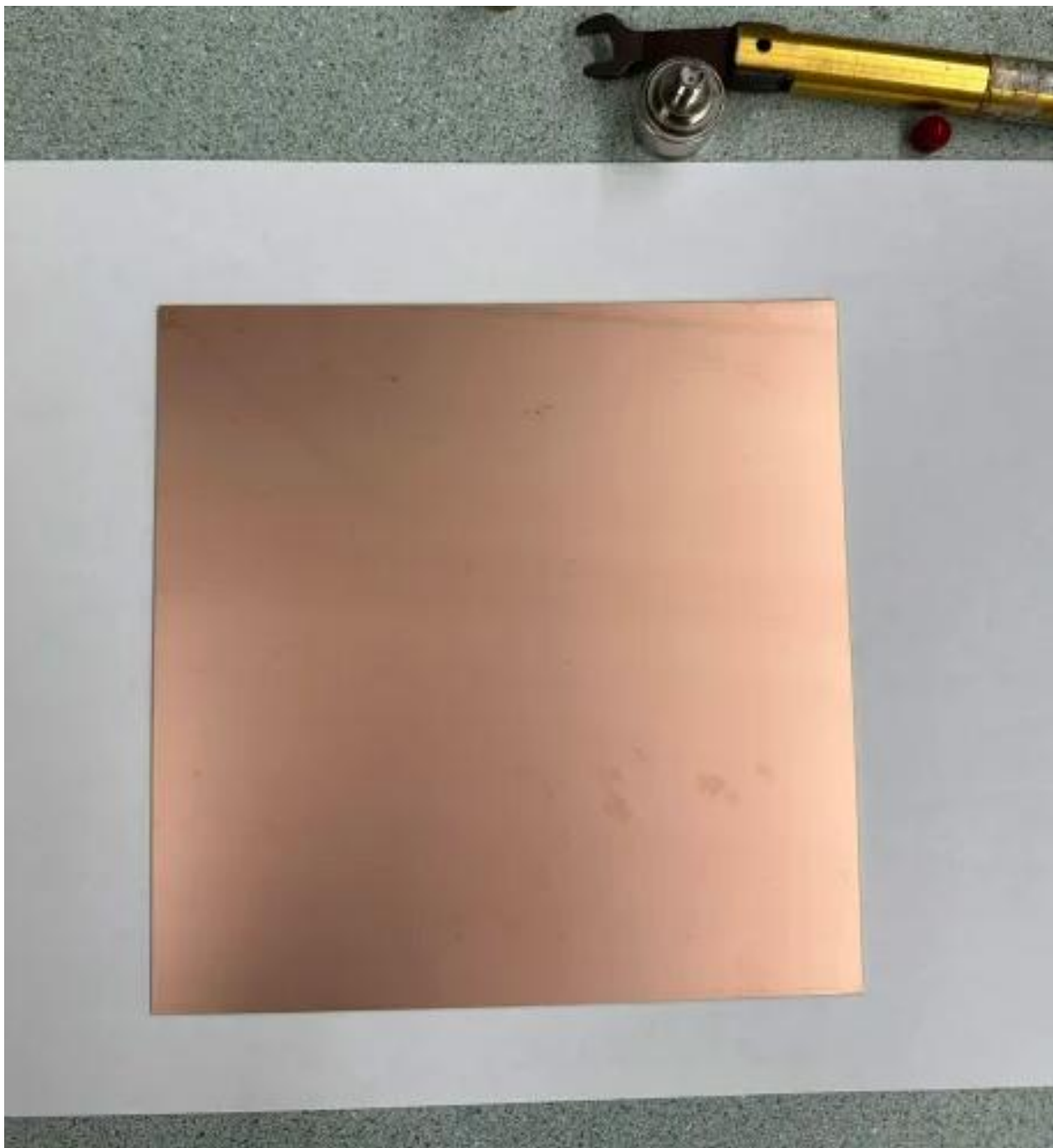
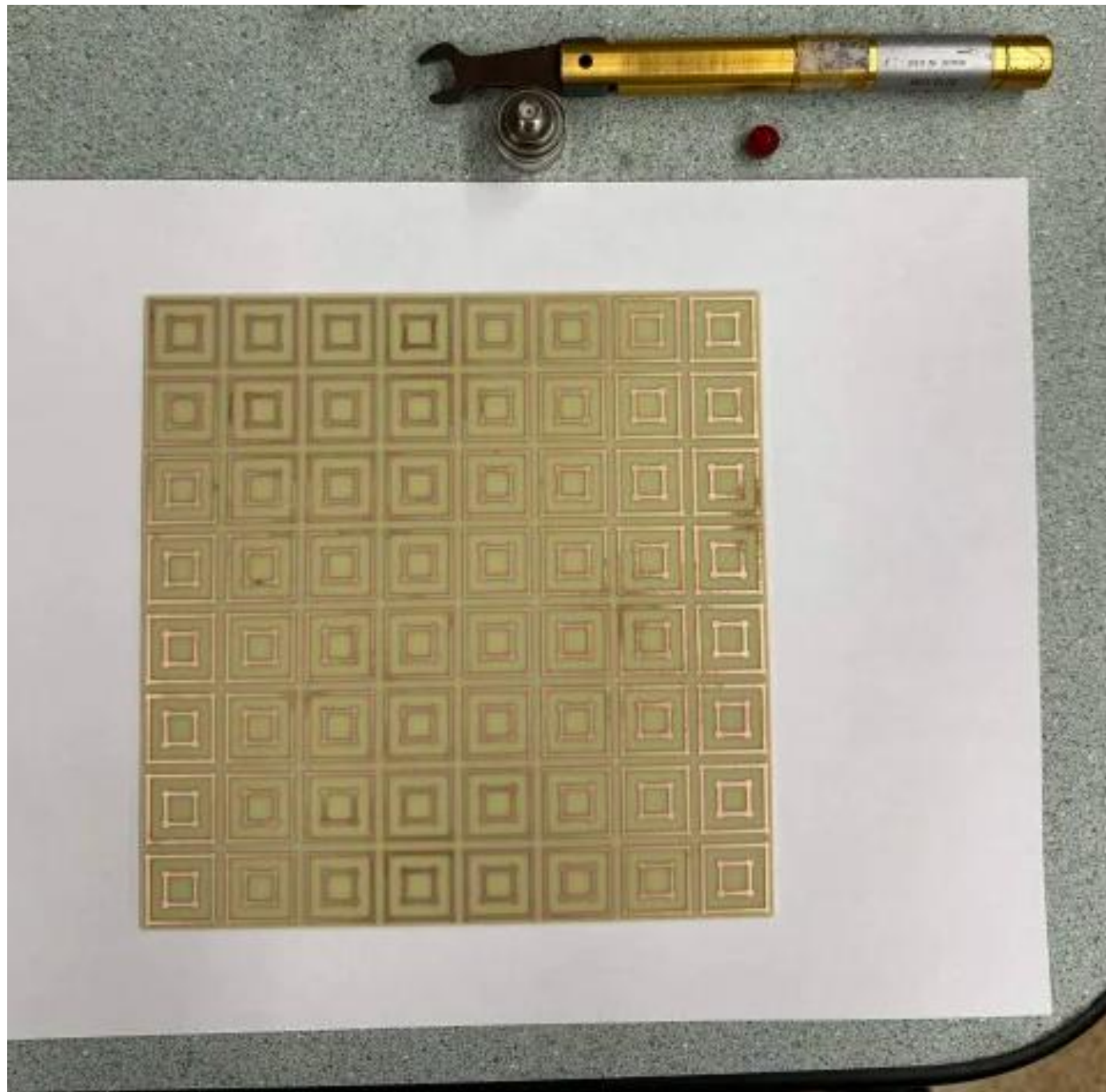
Kicad generation of gerber files

Design



Fabrication

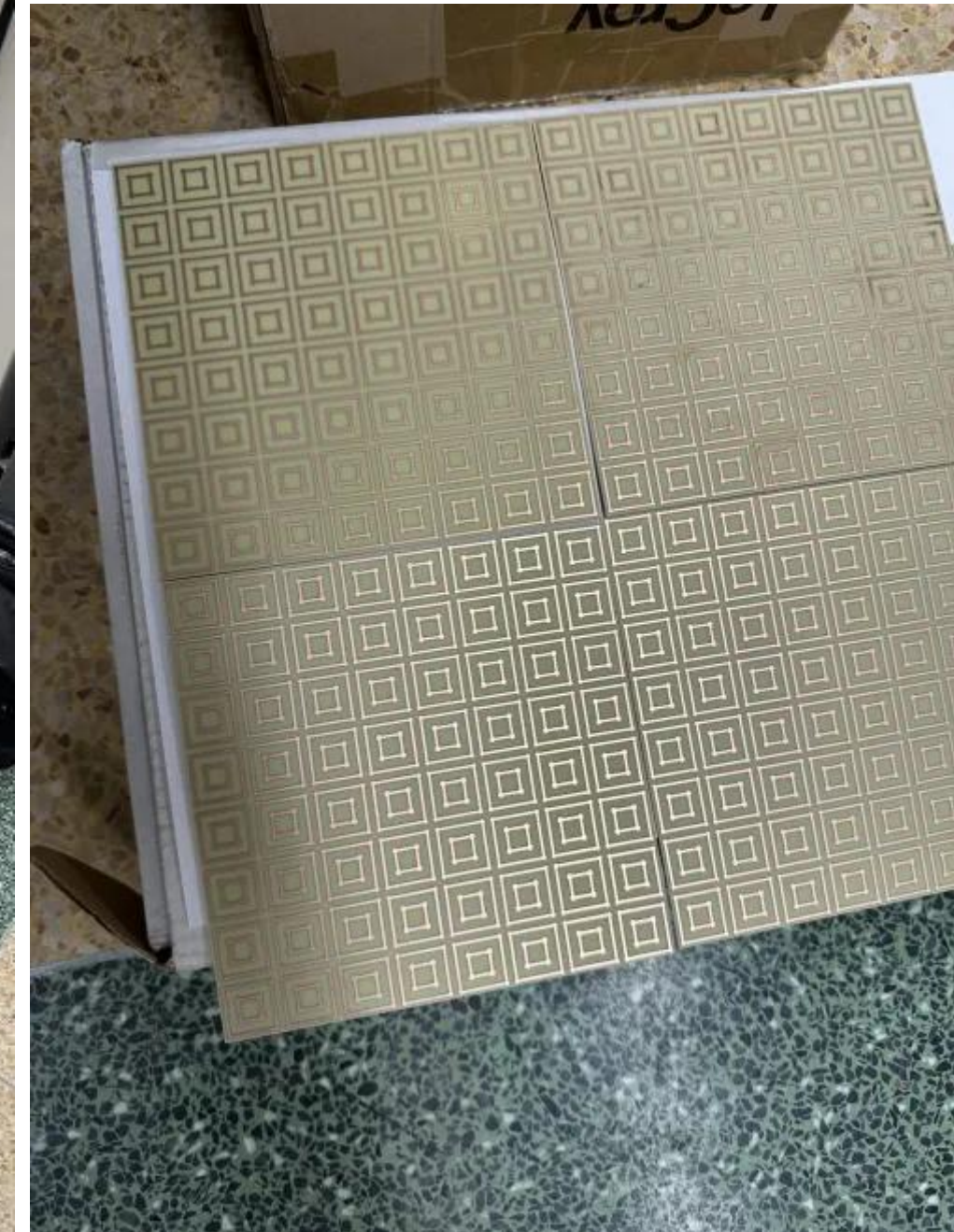
Testing



Setup

Testing

- The lack of an ideal environment poses challenges to accurate testing
- Without the presence of a surrounding such as a Faraday cage + anechoic chamber, unwanted signals and behavior will permeate the environment
- We calibrate our HP 8510B antenna and use a log-periodic antenna to produce S11 measurements
- For ease of setup and beamwidth coverage, we aim the antenna towards the ground, and cover the effective surface with multiple panels of our fabricated metamaterial



Setup

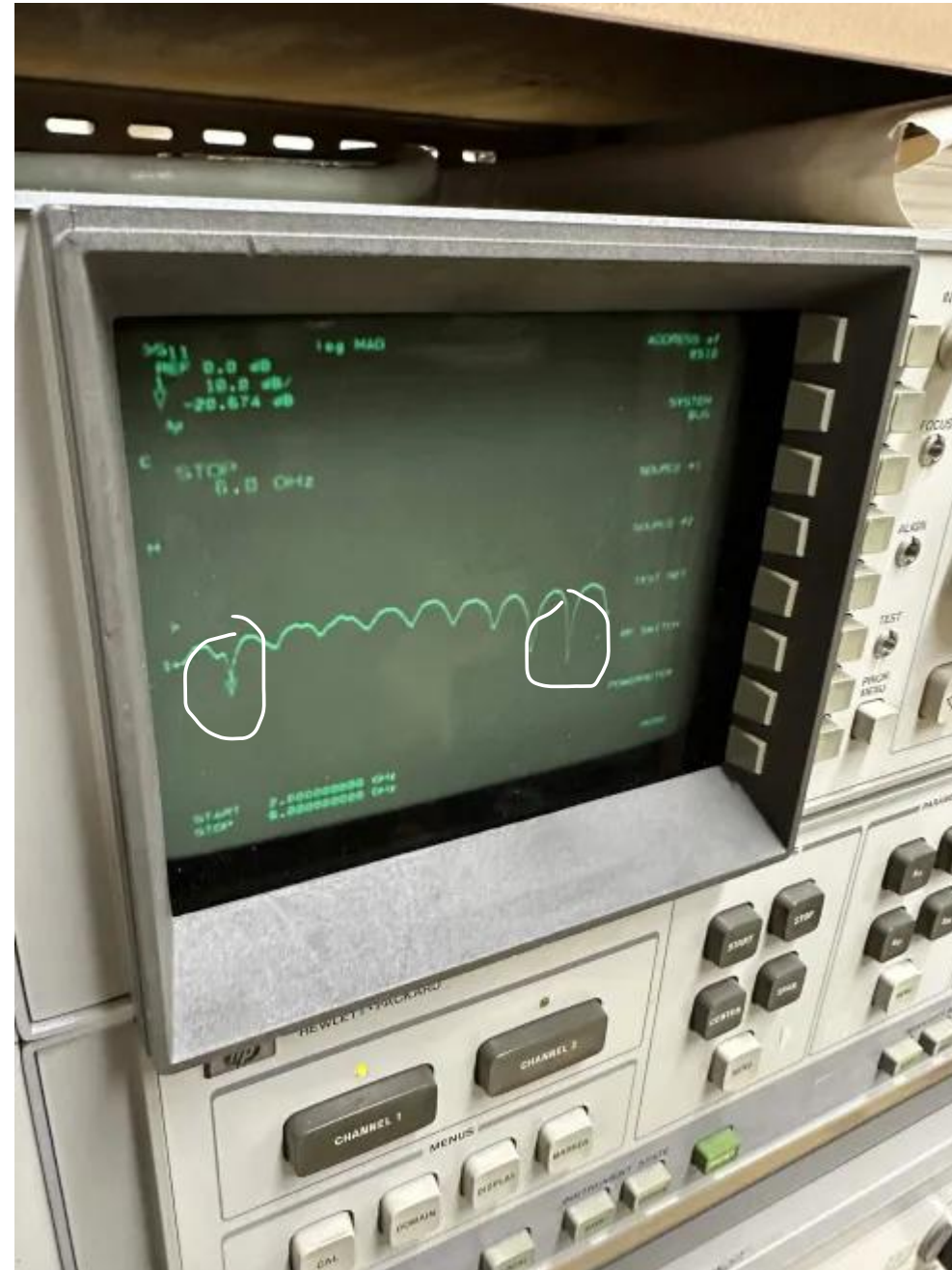
Testing

- Given that the far-field distance is different for 2.4GHz and 5.7 GHz, we set up two different heights, and run the network analysis on a local range with those two frequencies
- $d = \frac{2D^2}{\lambda} = \sim 56 \text{ cm @ } 2.4 \text{ GHz} \text{ \& } 103 \text{ cm @ } 5.7 \text{ GHz}$
- One full sweep scan at 103 cm distance away was also completed to show non-idealities of measurement
- Measurements were given in the form of complex numbers; python was used to extract mag S11 measurements



Results

Testing



Results

Testing

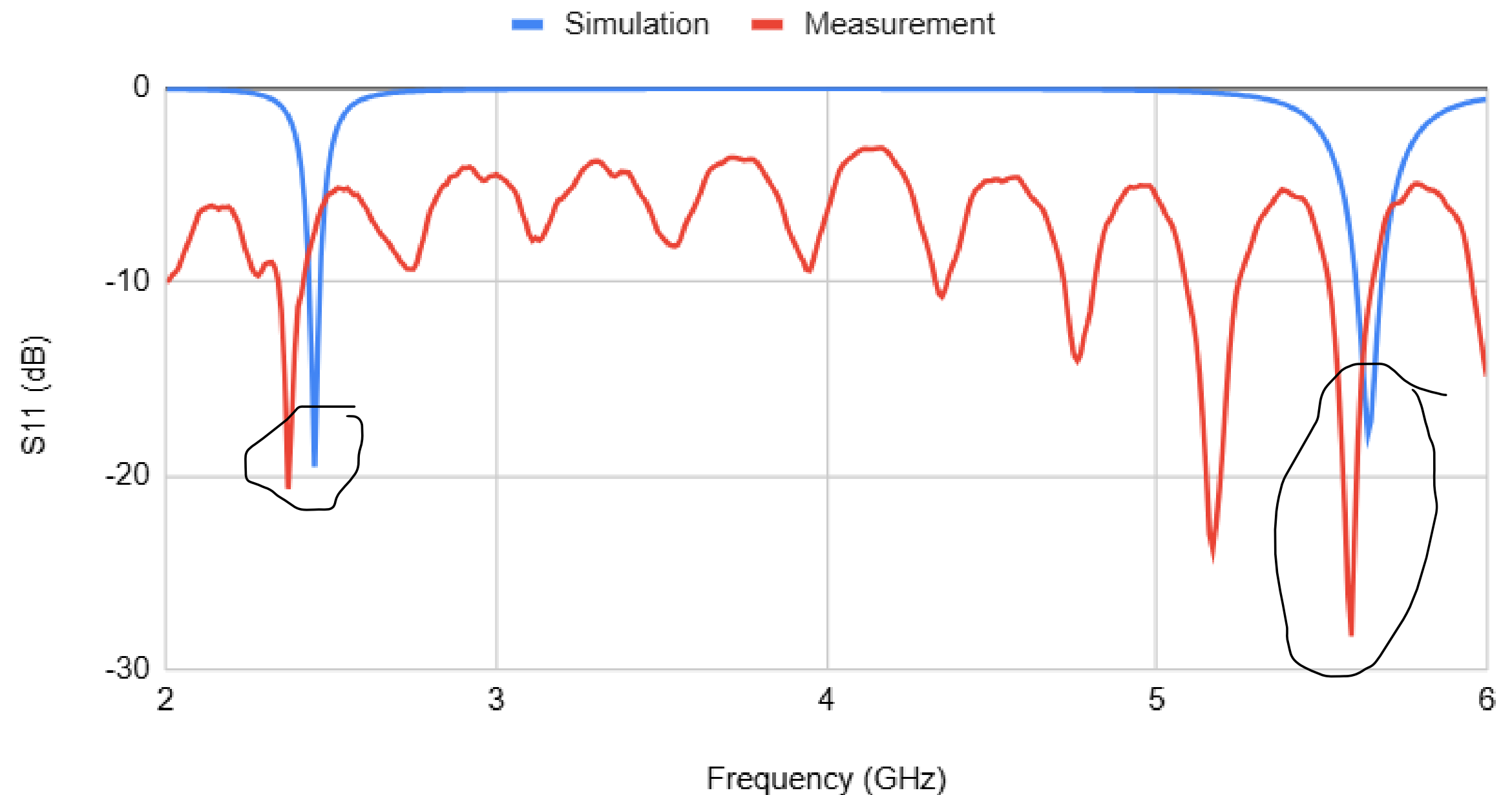


Simulation vs Measurement

Testing

- Given the non-ideal environments, we expect big discrepancies in frequencies that are not of interest
- However, we are able to capture the performance in the frequencies of interest due to our unique setup
- Differences in material properties from simulation to measurement strongly contribute to the noticeable frequency shift at the desired frequencies
- Eg. Peaks at 2.378 instead of 2.4 GHz, 5.634 instead of 5.7 GHz

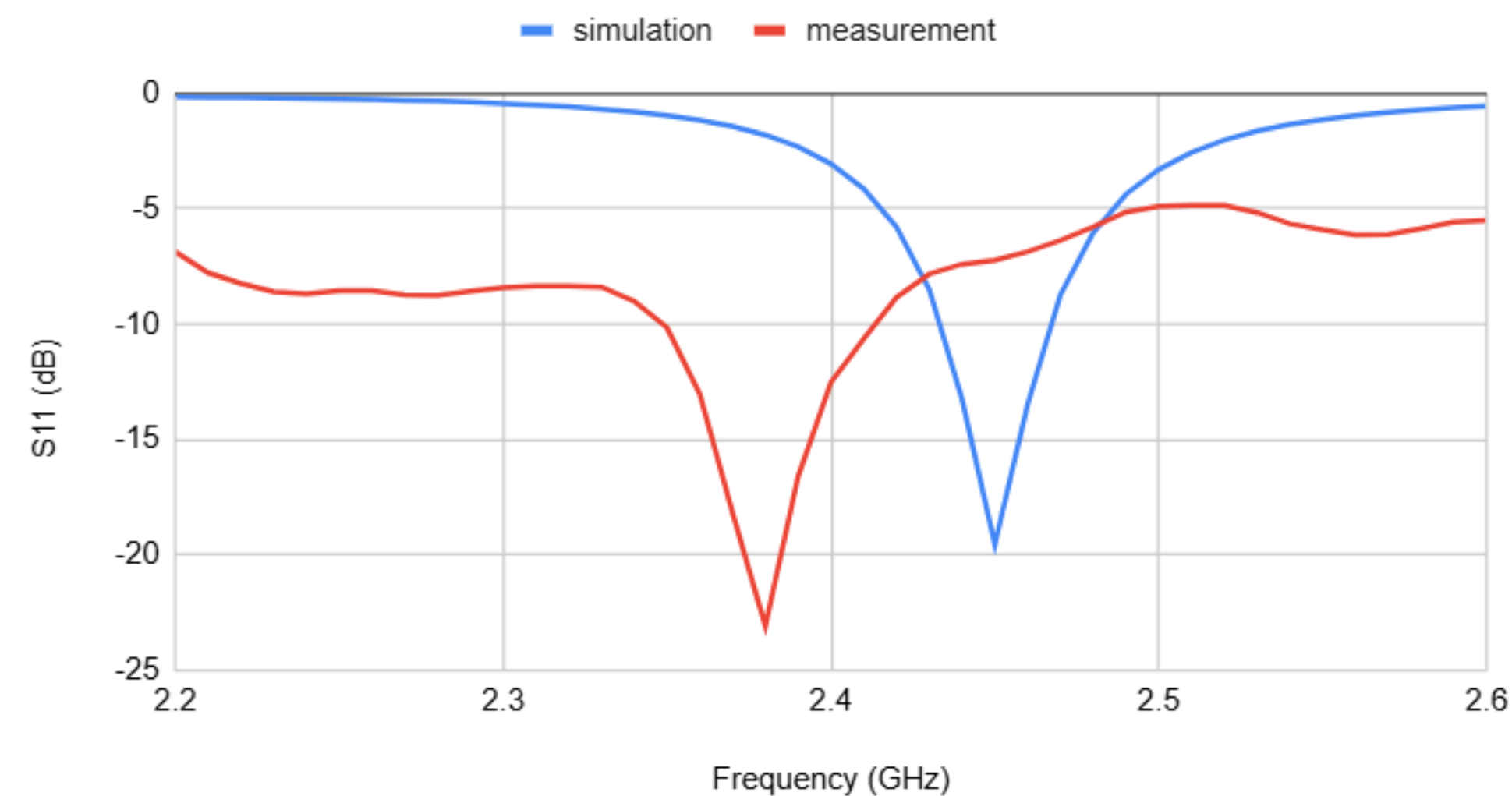
Simulated vs measured S11 results for 8x8



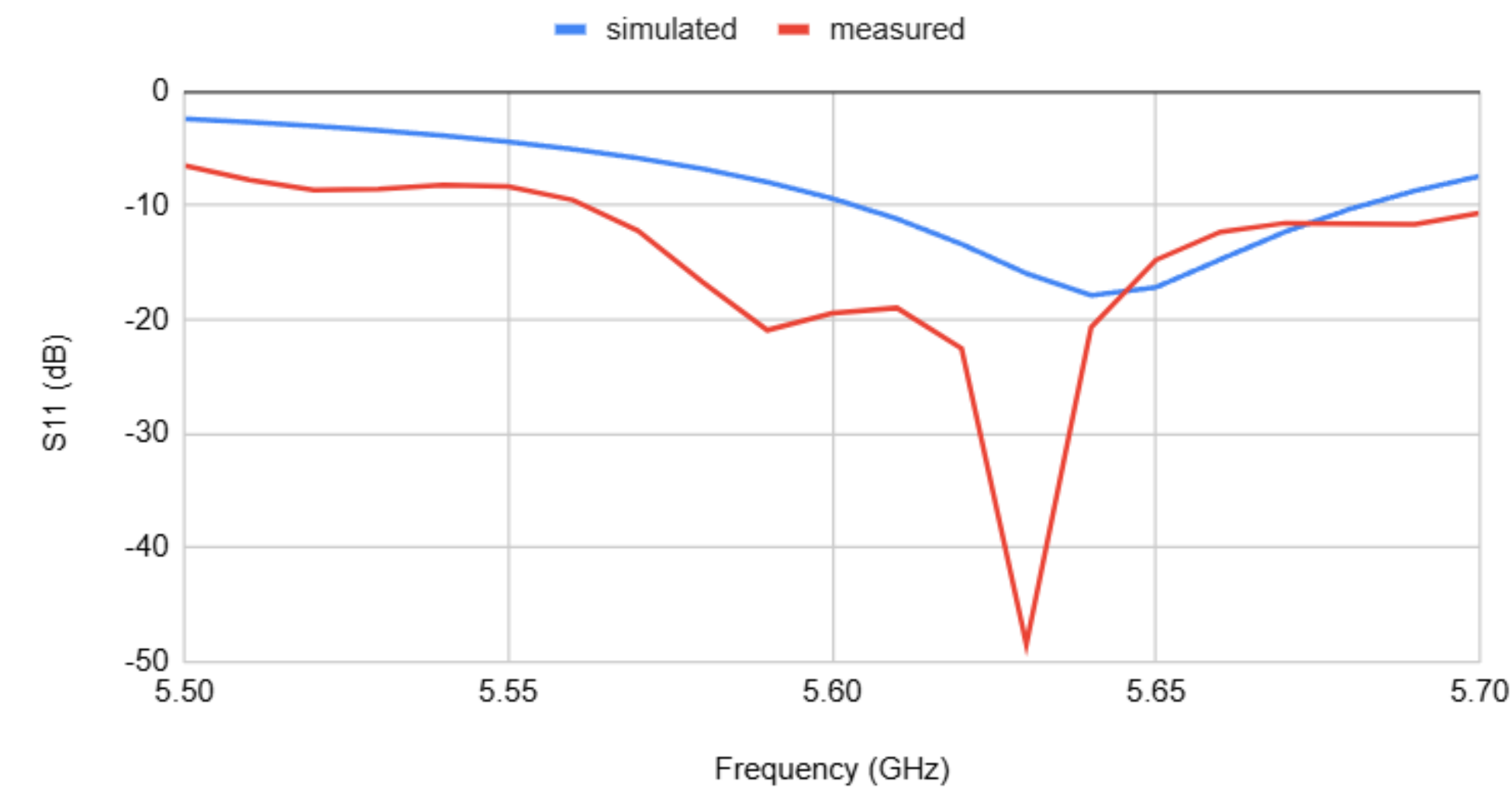
Local performance

Testing

2.4 GHz performance measured vs simulated



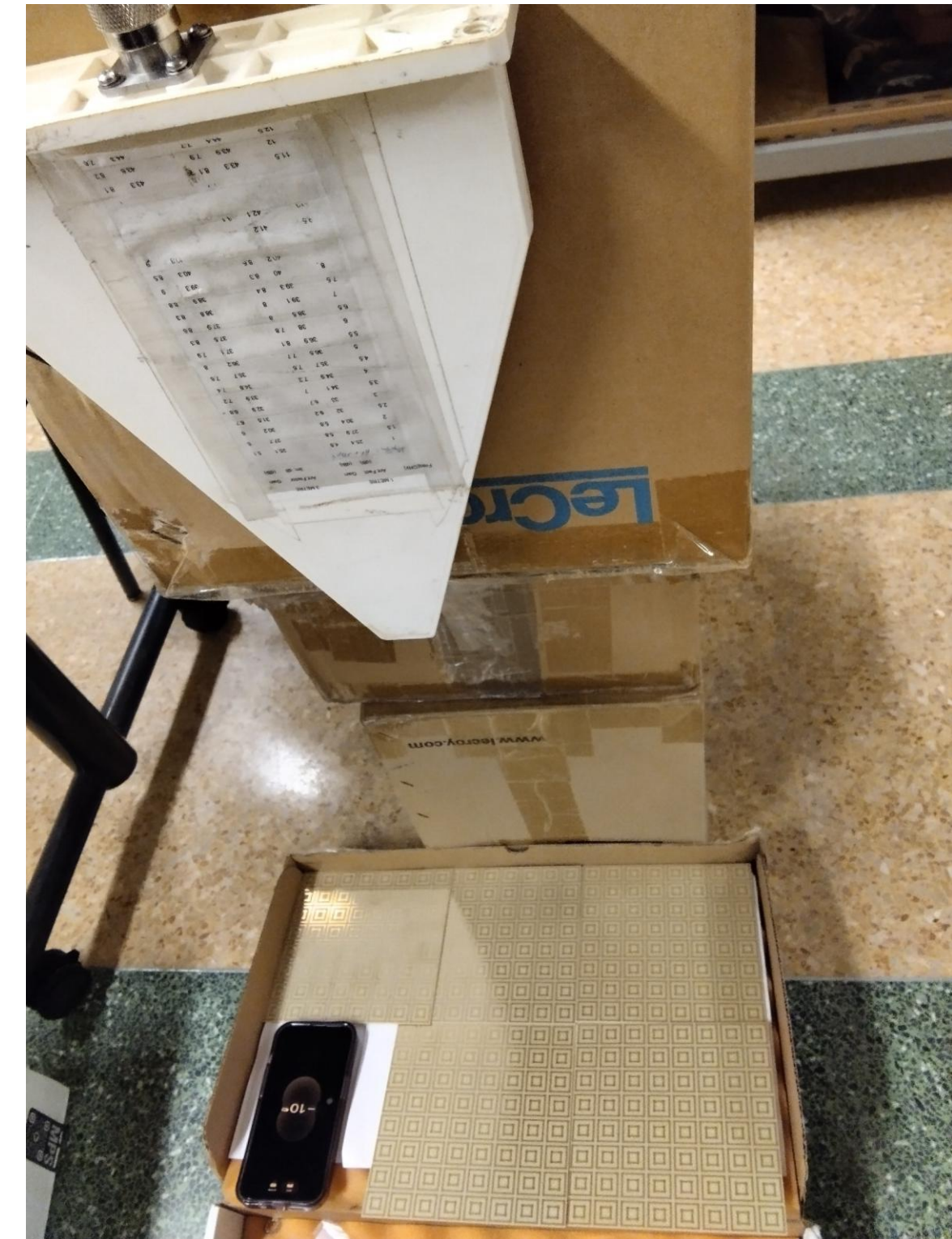
5.7 GHz performance measured vs simulated



Angled measurements

Testing

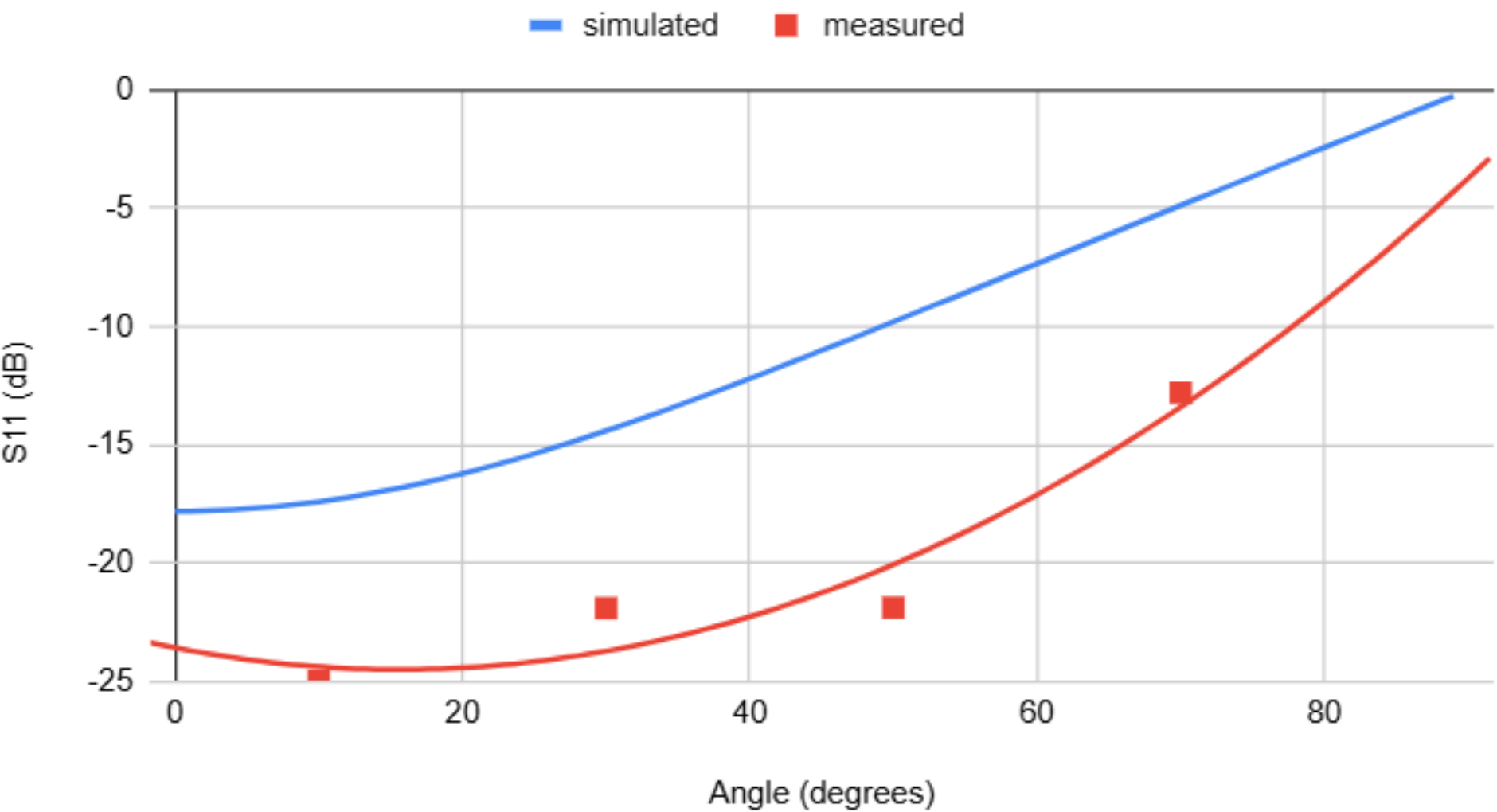
- To implement measurements that calculate S11 based on angle changes of the board, we set our boards at predetermined inclines that are verified using a compass app
- Both “roll” and “pitch” measurements were recorded to acquire TE and TM mode performance
- The “peak performance” across each measurement (from 10, 30, 50, 70 degrees) will be plotted against simulation



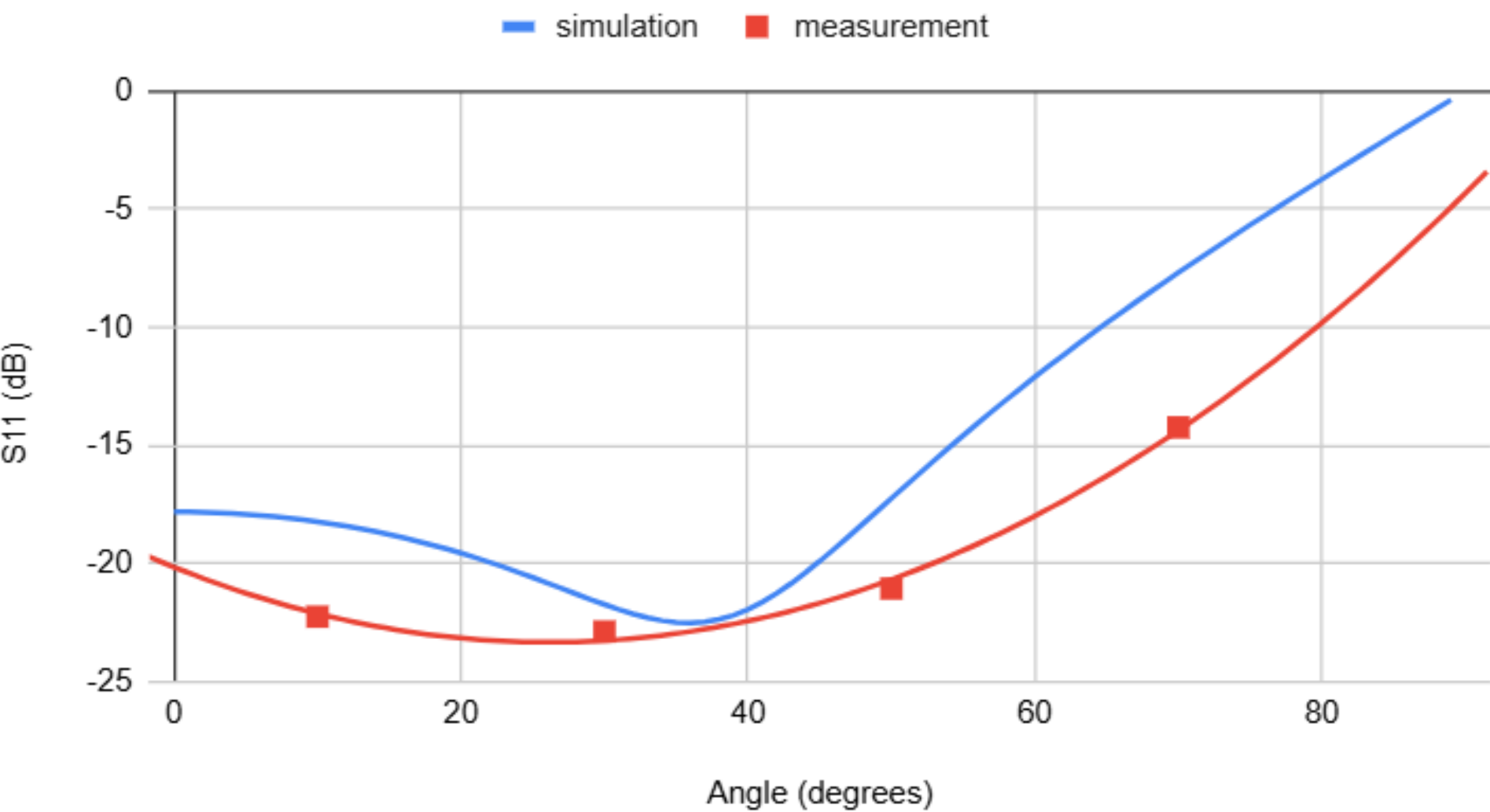
Angle measurement samples

Testing

TE mode simulated vs measured



TM mode simulation vs measurement



Discussion

Conclusion

- Although the device shows promising results in the absorption of RF energy, further improvements can be made through the inclusion of a via to another port, as shown in simulation
- Such a fabrication can reveal 2 port performance and can characterize the harvesting efficiency of the device
- This design used simple geometry and was low cost (using FR4), but would be quite lossy in real world applications (high loss tangent)
- It is expected to use RF materials (such as Rogers 4350B) in future designs